

Eberhard Karls University of Tübingen

Master's Thesis

The Formation of ExoMoons by Collisions

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MSc Astro and Particle Physics

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Abstract

The formation of moons is widely regarded as a natural outcome of planet formation, implying that exomoon satellites orbiting planets beyond our Solar System could be common, although none have been confirmed observationally. Detecting these moons remains challenging, yet current transit techniques can identify satellites larger than approximately 0.1 Earth mass. This study examines the collisional formation of exomoons through giant impacts between planets, using high-resolution simulations with 1 million particles to capture the dynamics of debris disk formation across a wide range of impact geometries.

This study focuses on collisions between planets of comparable size, with grazing angles of 15° , 30° , 45° , and 60° and the planets are moving at speeds that are close, to the speed they need to escape from each other. Two representative scenarios are considered: a $12 M_\oplus$ target struck by a $2 M_\oplus$ projectile (“Optimized Super-Earth”) and a $20 M_\oplus$ target impacted by a $5 M_\oplus$ projectile (“Giant Impacter Limit”). Simulations performed using the smooth particle hydrodynamics (SPH) method show that such giant impacts can generate extensive circumplanetary disks. Moons are usually formed from these disks away from the planets synchronous radius. This helps them move outward into orbits without getting destroyed.

Single big hits rarely create exomoons that we can detect with our tools. If there are many hits or hits that happen one after another, they could create bigger and more massive satellites. How fast the hit and at what angle it hits are very important. They control how much mass is in the disk and if a moon is formed. This shows that extreme hits are very important for making exomoons. These results help us understand what conditions are best for making exomoons that we can detect. They also guide us for looking for exomoons and, in the future.

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Chapter 1

Introduction

1.1 Background

During the past 30 years, the discovery of thousands of exoplanets and this has really changed how we think about planetary systems (Dressing and Charbonneau, 2013; Morton and Swift, 2013). At the time people have started to think about the possibility of exomoons, which are like moons that orbit planets outside of our own Solar System. From a point of view, it makes sense that moons would form when planets grow. If moons are present, they could affect how stable a planet's system is and whether or not the planet can support life. Despite what we would expect from theory we have not found any exomoons for sure yet. We know that if a moon is big enough 0.1 times the mass of Earth we should be able to see it using certain methods like watching how it blocks light from its planet (Kipping et al., 2012). However it is likely that making such big moons would require extreme and unusual events. One way that big moons might form is through collisions, which is what probably happened to make the Moon-Earth and Charon-Pluto (Canup and Asphaug, 2001; Canup, 2005). Recently, (Malamud et al., 2020) showed that big exomoons could form around certain types of planets but only if the collisions happen in very specific ways. I think we should look at all the way that exomoons could form not just the giant impacts. I should consider exomoons formation, in general, before focusing on impacts only and see how exomoons formation can happen in different scenarios.

1.1.1 Mass of the Debris Disk

The size of a moon that is made when two planets crash into each other depends on a thing. It depends on how much stuff goes into orbit around the debris (M_o), angular momentum (L_o), and how well the fragments of the debris stick together to form moons (Ida et al., 1997; Canup, 2008). These things about the disks around

the planets are affected by how the planets hit each other how fast they are going when they hit and what they are made of.

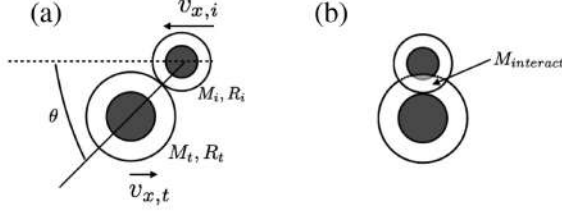


Figure 1: (Barr and Syal, 2017) (a) The diagram shows what happens when one planet (the impactor) hits another (the target). Both are made of rock (white) and metal (grey). The impactor has mass M_i and radius R_i , and the target has mass M_t and radius R_t with the angle of the collision is called θ . (b) The second part shows M_{interact} , which is the amount of material in the overlapping, lens-shaped area where the two bodies collide.

As shown in Figure 1(a), the geometry of the collision defines the impact angle, θ . Large disks capable of producing massive satellites typically form when impacts occur near the mutual escape velocity of the two bodies, i.e.

$$v_{\text{esc,sys}} = \sqrt{\frac{2G(M_t + M_i)}{R_t + R_i}}, \quad (1.1)$$

where M_t and M_i are the masses of the target and impactor, and R_t , R_i are their radii and G is the gravitational constant. The contribution to the impactor system is described by the mass ratio as

$$\gamma = \frac{M_i}{M_i + M_t}. \quad (1.2)$$

The disk mass is associated with the lens-shaped overlapping region of the colliding bodies with mass M_{interact} in Figure 1(b). Simulations suggest that for Earth-like targets, intermediate γ (0.1–0.2), and near-escape velocities, the scaling holds (Canup, 2008; Leinhardt and Stewart, 2012):

$$\frac{M_o}{M_T} \sim C_\gamma \left[\frac{M_i - M_{\text{interact}}}{M_T} \right]^2, \quad (1.3)$$

where $C_\gamma \sim 2.8(0.1/\gamma)^{1.25}$ is an empirically determined constant. If this expression applies broadly, it provides a way to estimate the masses of the disc and moon without full simulations. However, C_γ has only in limited cases, so its general reliability is uncertain.

1.1.2 Formation and Growth of Moons

After impact, the ejecta spreads rapidly onto a disk (Ward, 2012). The outer edge extends beyond the Roche limit as given by

$$a_R = \left(\frac{3\rho_{\text{pl}}}{\rho_s} \right)^{1/3} R_{\text{pl}}, \quad (1.4)$$

where ρ_{pl} and ρ_s are the densities of the planet and the disc material, and R_{pl} is the radius of the planet (Murray and Dermott, 1999; Elser et al., 2011). For Earth, this corresponds to $\sim 2.9R_{\oplus}$, with $R_{\oplus} = 6371$ km. Within the Roche limit, the material exists as a hot liquid–vapor mixture (Thompson and Stevenson, 1988; Ward, 2012; Charnoz and Michaut, 2015), while the disc evolves inward and outward depending on the thermal balance. At present, only upper estimates exist for possible moon masses. Collisions involving more massive planets create hotter and more vapor-rich discs, whose behavior remains poorly understood.

Many models assume that the disk condenses rapidly into solid particles (Ida et al., 1997; Kokubo et al., 2000; Hyodo et al., 2015). In such a particulate disk, gravitational dynamics dominates. For discs with $M_o/M_{\text{pl}} > 0.03$ (as in the Earth–Moon case), simulations favor a single moon. Its mass can be estimated as (Barr and Syal, 2017)

$$\frac{M_L}{M_o} \approx a \frac{L_o}{M_o \sqrt{GM_{\text{pl}} a_R}} - b - c \frac{M_{\text{esc}}}{M_o}, \quad (1.5)$$

where $a = 1.9$, $b = 1.15$, and $c = 1.9$. The mass lost from the system, M_{esc} , is less than 5%.

For intermediate discs ($0.003 < M_o/M_{\text{pl}} < 0.03$), N -body simulations predict that two moons may form, especially in low-angular-momentum cases (Hyodo et al., 2015). Their masses scale with M_o^2 , and they may settle into resonances such as 2:1 or become co-orbital (Kokubo et al., 2000; Hyodo et al., 2015).

For very small discs ($M_o/M_{\text{pl}} < 0.003$), simplified models predict (Crida and Charnoz, 2012)

$$\frac{M_L}{M_{\text{pl}}} \sim 2200 \left(\frac{M_o}{M_{\text{pl}}} \right)^3 \quad (1.6)$$

Only a few studies have combined evolution coupled with thermal disk evolution with subsequent accretion (Salmon and Canup, 2012). Their results for the canonical Earth–Moon impact suggests somewhat smaller satellites, with fitted parameters $a = 1.14$, $b = 0.67$, and $c = 2.3$. These values also hold for hotter and more massive discs formed in head-on impacts of bodies $\sim 0.5M_E$.

1.2 Motivation

The search for exomoons which are natural satellites that go around planets outside our Solar System is exciting in planetary science and astrophysics. Exomoons are interesting to researchers because they might affect the environments of planets and maybe even support life. Finding and confirming exomoons is very hard now which makes this area of study both rewarding and technically tough. One reason we want to study exomoons is that they might help make planets suitable, for life. Our Moon helps keep Earth's axis steady which gives us predictable seasons. Exomoons might do something for other planets and that's why researchers are studying them. The study of exomoons is important because it can help us learn more about planets and their moons.

Exomoons and planets are connected in ways and studying exomoons can help us understand these connections better. (Heller and Barnes, 2013) suggest that big exomoons could do the thing for planets outside our Solar System. By keeping the planets axis from changing much a moon could help prevent big changes in the climate. Without this stability a planets climate might become very harsh and unstable which would make it hard for life to exist. Exomoons can also affect what is going on inside planets through interactions. The gravitational pull between a planet and its moon can create forces that produce friction and heat inside the planet. We see this happening in our Solar System. For example, Saturn's moon Enceladus has oceans under its surface that are kept warm by energy. (Sasaki and Barnes, 2014) have shown that tidal heating can strongly influence the evolution of planets. In some cases, tidal energy might even create conditions that are warm enough to support liquid water on the surface of planets that are far from their stars. Finding exomoons would be a deal because it would help us understand how planetary systems form and evolve. By looking at how exomoons occur how big they are and what kinds of planets they orbit researchers can learn a lot about the processes that shape the architecture of planetary systems. This knowledge can also help us figure out how common it is for planets to have environments that can support life. However, forming big detectable exomoons might require specific conditions. Theoretical studies show that factors like the angle of impact the speed of collision and the mass of the objects involved are critical in forming moons. Barr and Syal (2017) and Malamud et al. (2020) have found that while big collisions can create debris disks around planets only a limited range of conditions leads to the formation of moons that are both big enough and stable over a long period of time. This suggests that big lived moons might not be very common because their formation depends on a narrow set of initial conditions. These constraints on formation affect how we search for exomoons. If detectable exomoons are rare and only form under circumstances, then we need to use careful modeling to identify which planetary systems are most likely to have big moons. Super-Earths, which

are planets that are bigger than Earth but smaller than gas giants are often studied in detail because they are one of the most common types of exoplanets found so far. By looking at how moons form around Super Earths astronomers can refine their search methods.

1.3 Objectives

This thesis aims to examine the formation of exomoons through giant- planetary collisions, employing smooth particle hydrodynamics (SPH) simulations. The primary goals are the following.

- Perform high-resolution SPH simulations with one million particles, allowing detailed tracking of the evolution of the debris disc after impacts.
- Analyze two representative collision scenarios:
 - the “Optimized Super-Earth” case ($12 M_{\oplus}$ target impacted by a $2 M_{\oplus}$ projectile), and
 - the “Giant Impactor Limit” case ($20 M_{\oplus}$ target impacted by a $5 M_{\oplus}$ projectile).
- Systematically looking at grazing angles 15° , 30° , 45° , and 60° , with velocities close to the planetary escape speed.
- Determine the impact conditions most conducive to producing circumplanetary discs capable of forming large, potentially observable moons.

1.4 Thesis Outline

The structure of this thesis is as follows.

- **Chapter 2 – Literature Review:** This chapter looks at the existing work on exomoons, including how to detect them how they form and what other studies have found. The thesis pays attention to Barr and Syal (2017), Canup et al. (2013), and Malamud et al. (2020). Section 2.1, Section 2.2 impact collisions and giant impact simulations, and Section 2.3 gaps in knowledge.
- **Chapter 3 – Methodology:** details the SPH approach used in this study, including the hydrodynamical code outline (Section 3.1), simulation setup (Section 3.2), numerical methods (Section 3.3), parameter choices (Section 3.4), and data analysis procedures (Section 3.5).

- **Chapter 4– Results:** presents simulation results. Section 4.1 illustrates a representative time evolution from impact to disc formation. Sections 4.2.1–4.2.6 analyze the exomoon candidates, mass distribution vs. grazing angle, trends in mass components, particle distributions for two scenarios, disk mass vs. grazing angle and particle count vs. grazing angle. Section 4.3 analysis of disk evolution (4.3.1-4.3.4), section 4.4 analysis of disk vapor fraction stability and comparative analysis of fragment stability and orbital architecture in tidal disruption events (Scenarios 1 and 2) (Sections 4.5.1-4.5.3).
- **Chapter 5 – Discussion:** interprets the results in the context of exomoon formation, stability, and detectability. Sections (5.1–5.3) discuss key interpretations, comparison with previous studies, and methodological limitations.
- **Chapter 6 – Conclusion and Future Work:** summarizes the main findings (Section 6.1) and describes potential avenues for future research (Section 6.2).

Chapter 2

Literature Review

2.1 Collisional Exomoon Formation

Exomoons are moons that move around planets outside our Solar System. These moons are interesting to scientists because they think they can form in ways. Some moons grow at the time as their planet. Some moons are caught when a planets gravity pulls in something that is passing by and some are made after a big crash.(Barr and Syal, 2017) shows that big crashes are how big rocky moons form around planets like Earth or big planets that are, like Earth. This happens when something big hits a planet. The crash sends pieces of rock and dust into space. After time, these materials are gathering and forms a moon. The size of the moon the way it orbits the planet and what the moon is made of all depend on how the objects were moving the angle of the crash and how big the objects were (Malamud et al., 2020). Looking at our Solar System helps us understand this process. For example, (Canup, 2004) how Earth's Moon could have formed from a crash and that is why the Moon is similar to Earth. (Canup et al., 2013) used high resolution simulations to show how the amount of momentum and the debris left in orbit affect a moon's final properties. These findings suggest that similar collisions could also form moons around planets outside our Solar System. Giant impacts are especially important for planets. This is because other methods like capture is less likely to happen under orbital conditions. A major collision does not create the debris needed to build a moon but also affects how the moon moves and spins. Simulations show that the angle and energy of the collision matter a lot. If a collision happens at an angle, it usually makes material go around the planet, which makes it more likely to form a moon. If the collision is very shallow or super-fast it might not leave pieces to make a moon (Barr and Syal, 2017; Malamud et al., 2020). Understanding these crash processes helps scientists guess where big exomoons might be and how we can find exomoons, around planets.

2.2 Impact Collisions and Giant Impact Simulations

Computer simulations are important for figuring out how moons form from crashes. We can use methods like SPH and adaptive mesh refinement (AMR) to see how the angle, speed and material distribution of a crash affect the moon that forms (Canup, 2004; Canup et al., 2013). Barr and Syal (2017) study looked at crashes on planets outside our solar system that are up to $10 M_{\oplus}$. They found that a few crashes made disks of debris that were big enough to form moons that we can detect.

These disks need to have a lot of mass around $0.3 M_{\oplus}$ to potentially form moons. The amount of material in the disk that forms a moon depends on how fast it's spinning and the physics of the disk. Some N-body models say that 10–55% of the disk material forms a moon (Kokubo et al., 2000), while complex models say it is around 20–45% (Salmon and Canup, 2012). When two big objects crash they create a disk of melted and vaporized rock that spreads out beyond the Roche limit, where the gravity of the planet becomes too strong. This is where the fragments of rock can come together to form moons (Ward, 2012; Charnoz and Michaut, 2015). There is a model called BB17 that scientists use to study how galaxies form and change. BB17 also analyzed how the disk-to-total mass ratio (M_o/M_T) correlates with the normalized angular momentum (J_{col}), calculated as

$$J_{\text{col}} = 2f(\gamma)\left(\frac{v_{\text{imp}}}{v_{\text{esc}}}\right)\sin\theta, \quad f(\gamma) = \gamma(1-\gamma)\left[\gamma^{1/3} + (1-\gamma)^{1/3}\right]^{1/2}, \quad (2.1)$$

where θ is the impact-angle, v_{imp} the impact-velocity, v_{esc} the escape velocity, and γ the impactor to the total mass ratio (Canup, 2005). Their results show that collisions with high velocity or very shallow angles are less likely to form massive moons, whereas more collisions of equal mass (higher γ) are favorable, similar to Pluto–Charon. Additionally, moons can grow through multiple impacts rather than a single event, interacting gravitationally, merging, or even falling back onto the planet (Rufu et al., 2017; Citron et al., 2018; Malamud et al., 2018). Tracking these processes with SPH and N-body simulations helps to estimate the final moon masses, compositions, and orbits around superterrestrial exoplanets.

2.3 Gaps in Knowledge

There is some progress in simulations and theoretical modeling and still a lot of things do not know. Most models are based on ideas about what planets are made of and what happens when they collide, but real planetary systems are probably

a lot more complicated. There are the collisions and changes over time (Barr and Syal, 2017; Malamud et al., 2020). The question of whether moons that form after big collisions can survive for a time is also not fully answered. Big planets, those that are a lot bigger than Earth can pull hard on the moons that orbit them which can change the moons path and even make it fall into the planet or get thrown out of the system. We do not have information about big rocky moons that orbit other planets, which makes it hard to test our models and figure out things like how big the objects that collide are how often they collide and what happens to the disk of material that forms after a collision. (Canup et al., 2013) have also pointed out that making a moon with the composition as our Moon requires very specific conditions. This probably means that making moons in other planetary systems also requires very specific conditions. To learn more, about this we will need to make our computer models more advanced and look closely at other planets to see if they have moons.

Chapter 3

Methods

3.1 Hydrodynamical Code Outline

The simulations in this study were conducted using the `miluphcuda` SPH code (Schäfer et al., 2016, 2020). It is a GPU-accelerated SPH implementation, achieving computational speeds up to 10–100 times faster than CPU counterparts, depending on the GPU architecture. This allows high-resolution simulations with up to 10^6 SPH particles, sufficient to capture the morphology of the debris disk and the accretion of clump.

The code implements a Barnes–Hut tree algorithm for efficient self-gravity calculations and supports a variety of physics, including elastic and plastic solids, fracture mechanics, and full hydrodynamics. For this study, I used the hydrodynamic mode only, neglecting solid-body rheology, in agreement with previous studies on Moon and exomoon formation (Canup, 2005; Barr and Sval, 2017; Malamud et al., 2020).

Thermodynamics is modeled using the Tillotson equation of state for granite and iron under extreme P – T conditions. Parameter files were adapted from (Melosh, 2007), consistent with canonical Earth–Moon formation models (Canup, 2005; Barr and Sval, 2017).

3.2 Simulation Setup

For each run, the simulation setup was prepared in a pre-processing stage. In this step, both the target and the impactor were constructed with internally relaxed structures. Their density followed hydrostatic profiles, and internal energy values were obtained through adiabatic compression. This semi-analytical method, which is consistent with the same physical relations applied in the SPH model, provides a much faster alternative to the lengthy approach of letting each body evolve on

its own into hydrostatic equilibrium, a method adopted in earlier works such as (Canup et al., 2013; Schäfer et al., 2016, 2020).

Together, eight simulations were performed on the bwForCluster BinAC at Tübingen University, using NVIDIA A100 80 GB PCIe GPU with CUDA 12.8.

Each job was executed on a single dedicated GPU and typically lasted 10–15 days, for two cases "The Optimized super-Earth" and "The Giant Impactor Limit", particle resolution of 10^6 SPH elements was employed, striking a balance between runtime and accuracy. However, it should be emphasized that the SPH method has known limitations in regions of low density (Reinhardt and Stadel, 2017). I looked at what happens when Earth and the Moon collide and when other planets crash into each other (Canup and Asphaug, 2001; Canup, 2005; Barr and Syal, 2017; Malamud et al., 2020), I came up with two ideas for how these impacts could happen:

Scenario 1 — Optimized Super-Earth

Target mass: $12 M_{\oplus}$, Projectile mass: $2 M_{\oplus}$

Scenario 2 — Giant Impactor Limit

Target mass: $20 M_{\oplus}$, Projectile mass: $5 M_{\oplus}$

Each scenario was simulated at grazing angles of 15° , 30° , 45° , and 60° . Planets were set up to be, like our Earth with iron cores and granite mantles.

Initial conditions:

Number of particles: $N = 10^6$

Initial rotation period: $P_{\text{rot}} = 0$

Initial separation: $r_{\text{sep}} = 3 \times R_{\text{touching}}$

miluphcuda command used:

```
time ./miluphcuda/miluphcuda \
-v -f impact.0000 -g -H -I rk2_adaptive \
-Q 1e-5 -m material.cfg -n 500 -t 100.0 \
-s > miluphcuda_run.log 2>&1
```

3.3 Numerical Methods

miluphcuda solves the hydrodynamic equations:

$$\frac{d\rho}{dt} = -\rho \nabla \cdot \mathbf{v}, \quad \frac{d\mathbf{v}}{dt} = -\frac{\nabla P}{\rho} + \mathbf{g}, \quad \frac{du}{dt} = -\frac{P}{\rho} \nabla \cdot \mathbf{v},$$

where ρ is the density, $\mathbf{v}$ is the velocity, P is the pressure, u is the specific internal energy, and $\mathbf{g}$ is the gravitational acceleration.

Hydrodynamics: Standard SPH formulation using a smoothing kernel, with artificial viscosity to capture shocks Schäfer et al. (2016, 2020).

Gravity: Self-gravity is computed efficiently using a Barnes–Hut tree algorithm.

Equation of State: The Tillotson EOS is used to model the granite and iron components. The Tillotson EOS is good for high-velocity impacts. It gives a simple yet effective description of material behavior under extreme pressures and energies. For compressed regions ($\rho \geq \rho_0$) and internal energy u lower than the energy of incipient vaporization u_{iv} , the equation of state (Tillotson, 1962; Melosh, 1996) is:

$$p = \left[a_T + \frac{b_T}{1 + u/(u_0\eta^2)} \right] \rho u + A_T \xi + B_T \xi^2, \quad (3.1)$$

where,

$$\eta = \frac{\rho}{\rho_0}, \quad \xi = \eta - 1. \quad (3.2)$$

For expanded material ($u > u_{cv}$), the EOS takes the form:

$$p = a_T \rho u + \left[\frac{b_T \rho u}{1 + u/(u_0\eta^2)} + A_T \xi e^{-\beta_T(\eta^{-1}-1)} \right] e^{-\alpha_T(\eta^{-1}-1)^2}. \quad (3.3)$$

The symbols $\rho_0, A_T, B_T, u_0, a_T, b_T, \alpha_T$, and β_T are material-dependent parameters. In the partial vaporization regime ($u_{iv} < u < u_{cv}$), the pressure is linearly interpolated between the two values obtained from the compressed and expanded forms, respectively.

The Tillotson EOS is computationally simple yet suitable for a wide range of high-pressure applications such as high-velocity collisions and impact cratering. It can yield negative pressures for expanded ($\eta < 1$) and cold ($u < u_{iv}$) states, corresponding to tension in the solid material. To address this, a density limit parameter can be defined for each material type in the simulation, once η falls below this limit, the pressure is set to zero. Some simulation codes also include a compile-time option to prevent negative pressures entirely.

Resolution: I am using $N = 10^6$ SPH particles, which is enough to see debris and start of clump formation and computationally feasible on GPU hardware.

Outputs: I recorded every 0.1 code units (~ 0.1 hours) and see the sequence: impact $\rightarrow$ debris ejection $\rightarrow$ disk spreading $\rightarrow$ satellite accretion. This setup follows the implementation described in (Schäfer et al., 2016, 2020), this means we can do detailed simulations, on modern GPUs and still get the important parts of what happens when super-Earths and big things collide.

3.4 Parameter Choices

Key parameters explored:

- **Mass ratios:** 6 : 1 (Scenario 1), 4 : 1 (Scenario 2)
- **Simulation Scenarios**

Table 3.1: Scenario 1 Parameters — The “Optimized Super-Earth” ($12 M_{\oplus}$ target + $2 M_{\oplus}$ projectile)

Desired Grazing Angle (°)	M_tot (kg)	M_proj (kg)	Impact Angle (°)
15	1.493×10^{26}	1.1944×10^{25}	75
30	1.493×10^{26}	1.1944×10^{25}	60
45	1.493×10^{26}	1.1944×10^{25}	45
60	1.493×10^{26}	1.1944×10^{25}	30

Table 3.2: Scenario 2 Parameters — The “Giant Impactor Limit” ($20 M_{\oplus}$ target + $5 M_{\oplus}$ projectile)

Desired Grazing Angle (°)	M_tot (kg)	M_proj (kg)	Impact Angle (°)
15	1.493×10^{26}	2.986×10^{25}	75
30	1.493×10^{26}	2.986×10^{25}	60
45	1.493×10^{26}	2.986×10^{25}	45
60	1.493×10^{26}	2.986×10^{25}	30

- **Thermal state:** The Tillotson EOS is a pressure–density–internal energy relation,

$$p = p(\rho, u).$$

This thing is different from more complex EOS like ANEOS or SESAME. It does not directly use temperature as a variable. The internal energy is what matters here. This internal energy includes everything that has to do with heat, such, as the way atoms vibrate or how hot something gets. So, the state of heat is basically part of the energy.

This space of parameters is a mix of being realistic and not using too much computational, which is what other people have done when they studied how exomoons form (Heller and Barnes, 2013; Malamud et al., 2020).

3.5 Data Analysis Procedures

3.5.1 Debris Disk Diagnostics

The Debris Disk mass was measured about 24 hours after something hit it. I looked at how the Angular Momentum and Energy were shared out. I figured out where the material came from whether it was from the target or the projectile.

3.5.2 Moon Formation Tracking

I used a special method to find clumps that were stuck together because of gravity and applied some rules to see if they were stable:

Roche Limit: The Roche limit is calculated as,

$$d_{\text{Roche}} = 2.44 R_p \left(\frac{\rho_p}{\rho_s} \right)^{1/3}, \quad (3.4)$$

where R_p and ρ_p are the radius and density of planet, and ρ_s is the density of the satellite. If clumps are too close to the planet inside this d_{Roche} they get broken apart by the planet's gravity.

Synchronous Radius: The synchronous radius is defined as,

$$r_{\text{sync}} = \left(\frac{GM_p P_{\text{rot}}^2}{4\pi^2} \right)^{1/3}, \quad (3.5)$$

where P_{rot} is the how long it takes the planet to rotate once, G is universal gravitational constant and M_p is the mass of planet.

Chapter 4

Results

4.1 Showcase Figure

Figure 2 presents a step-by-step visual overview of the “Optimized Super-Earth” simulation at grazing angle 45° for Scenario 1. The sequence of images shows how a moon can form after a collision between two large bodies. A moon can form after a collision between two large bodies. The first stage is when the two objects crash into each other at high speed and start of how a moon can form after a collision, between two large bodies. Now, it releases a huge amount of energy and breaks parts of both bodies. Hot, melted material and rock are thrown out into space. The planets are not separate anymore they are now one thing with a lot of junk around it. After they crashed into each other the gravity and the way the main body spins around start to affect the junk that is floating around. Some of the junk falls back to the body but a lot of it does not. The main body spins around fast and this makes the junk go out in all directions and start to go around the main body. The junk forms a curve of stuff that starts to look like a disk and this disk is made of rocks and dust. The main body and the disk of rocks and dust are what we have now. The disk is important because it is made of the same stuff, as the main body. This disk contains the material that can later form moons. As time passes, the debris disk becomes flatter and more organized. Inside this disk, small pieces collide with each other and stick together. This process is called accretion. Gradually, larger clumps form as they collect more material. These growing bodies begin to clear their paths by gathering nearby dust and smaller fragments.

In the final stage, the simulation shows the result after all the material has come together. The system is now calm and steady and big central body is the thing we can notice. Most of the junk that was floating around has. Gone back to the central body or drifted away and some of the bigger pieces have stuck together

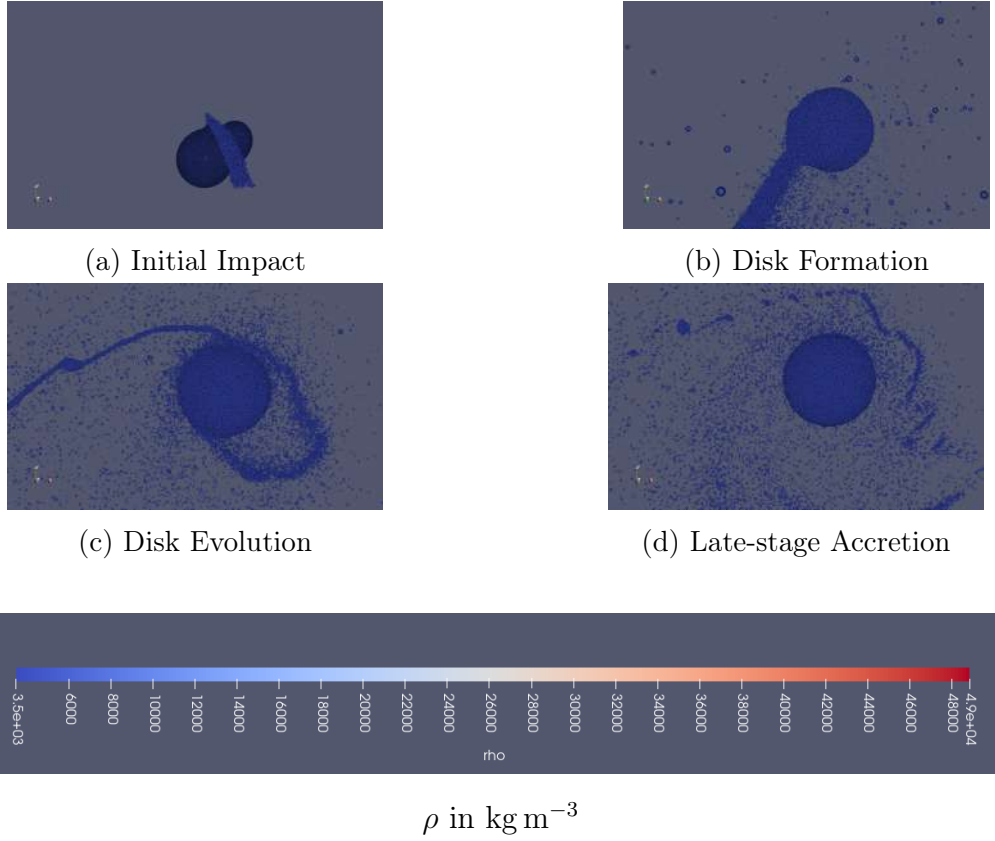


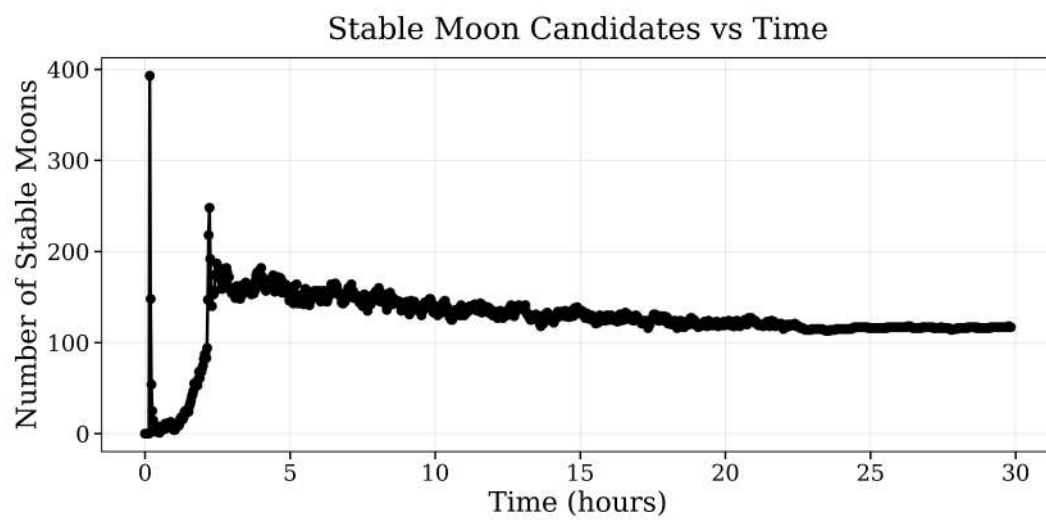
Figure 2: Representative simulation time evolution showing the sequence from initial impact (Panel a) to late-stage accretion (Panel d).

to make a moon. The system is no longer wild and crazy it has become stable. The stable system has really taken shape. The central body is the main object, in this stable system. This shows how a powerful collision can change a messy and violent event into an organized system, with a planet and a moon orbiting around it.

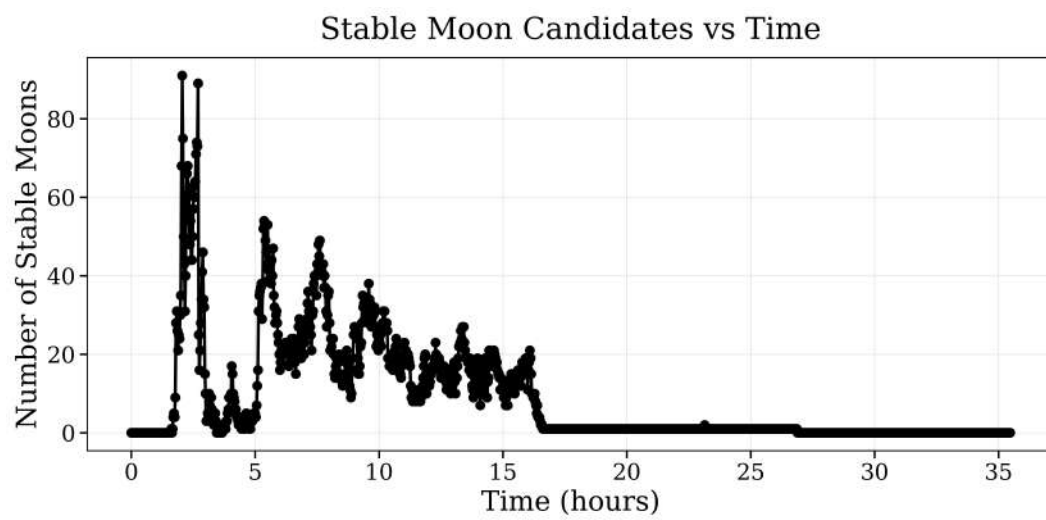
4.2 Core Results

4.2.1 Exomoon candidates vs. Time

The number of exomoon candidates and the way they behave change a lot over time. We can see this in Figures 3 and 4 for two simulation scenarios, S1 and S2. Each scenario was looked at with grazing angles: 15° , 30° , 45° and 60° . At the start many fragments are thought to be possible exomoon candidates. This is because I use a friend-of-friends algorithm that groups particles that are close to each other in space. The friend-of-friends algorithm links any pair of fragments that are within a certain distance of each other, in the friend-of-friends algorithm. This way I can find groups of fragments that are connected by gravity without assuming what shape they are. The friend-of-friends process finds all areas that are densely packed and could possibly form bound groups. Once I find these groups and use criteria to select stable exomoon candidates. I only keep fragments that have orbital energy relative to the planet are outside the Roche limit and are within the Hill radius. This ensures that only fragments that are connected by gravity and are stable are counted as exomoon candidates. In all scenarios a lot of candidates are lost in the 10 to 15 hours. This shows how important tidal forces and gravity are in shaping the orbits of exomoons from the beginning. When the grazing angle is 15° the simulations start with candidates, but they disappear quickly. At angles of 30° and 45° the number of candidates decreases slowly so some fragments can survive for longer. When the grazing angle is 60° the simulations consistently have the lowest long-term stability with almost no candidates left after a while even if there are a lot at the start. Overall, these results show that exomoon candidates do not survive for long and are very sensitive to the grazing angle of the impact. Under certain conditions can exomoons exist for a long time. The evolution of all fragments after filtering are summarized in Table 4.10. Exomoon candidates and their behavior is important to understand. Exomoon candidates are affected by the grazing angle and the time that passes. Exomoon candidates are a part of this study.



(a)



(b)

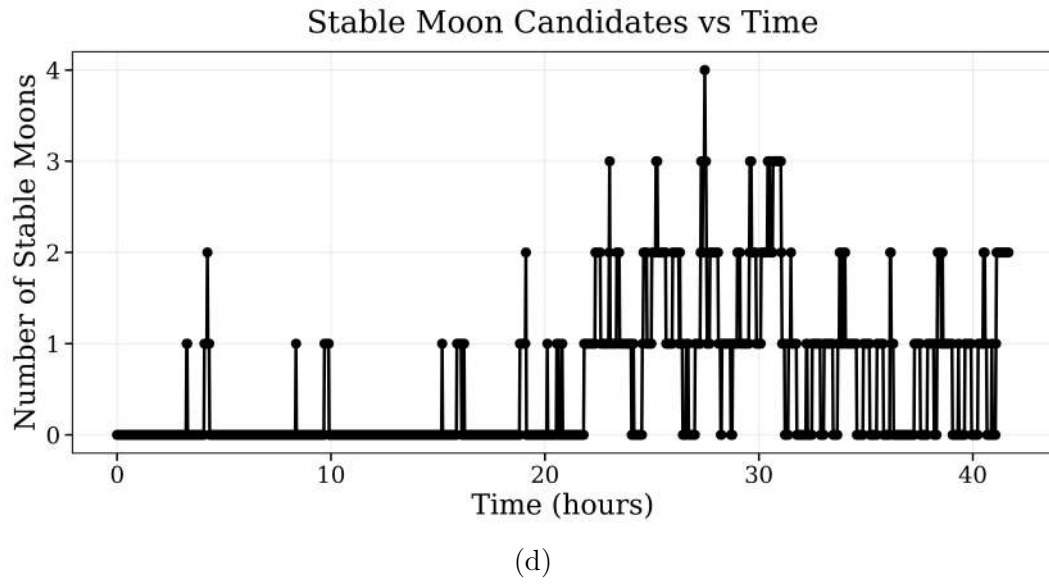
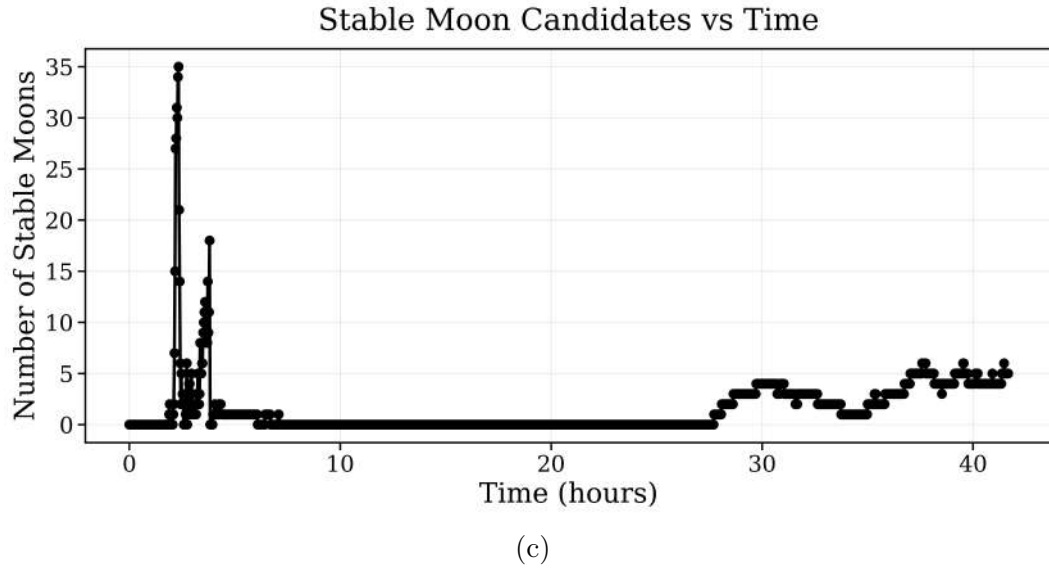
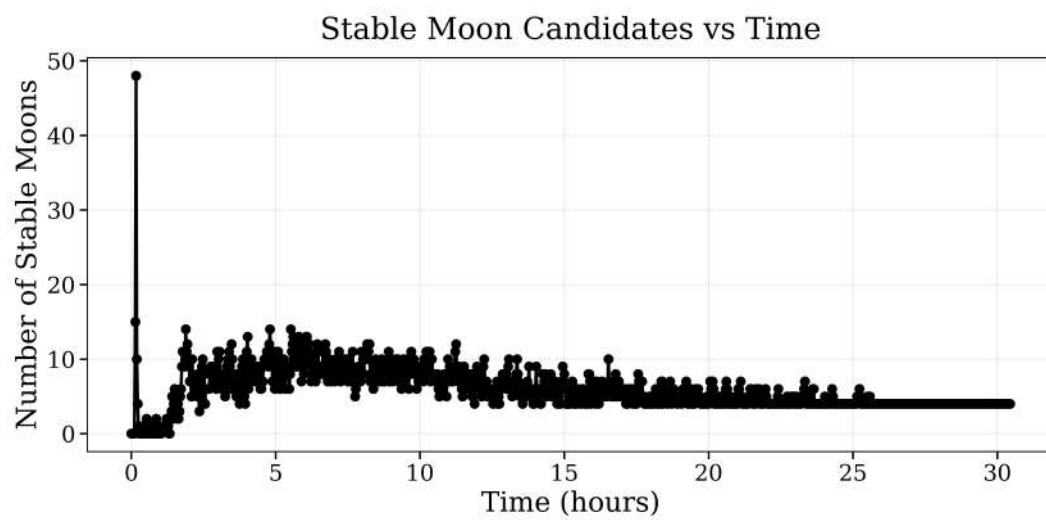
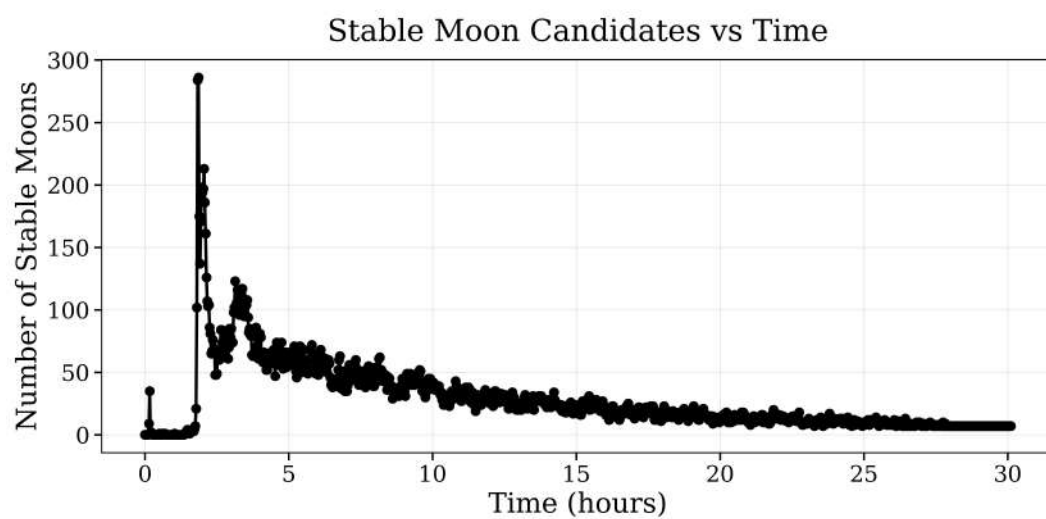


Figure 3: (a), (b), (c), and (d) shows the time evolution of stable exomoon candidates for simulation S1 at grazing angles 15° , 30° , 45° , and 60° , respectively.



(a)



(b)

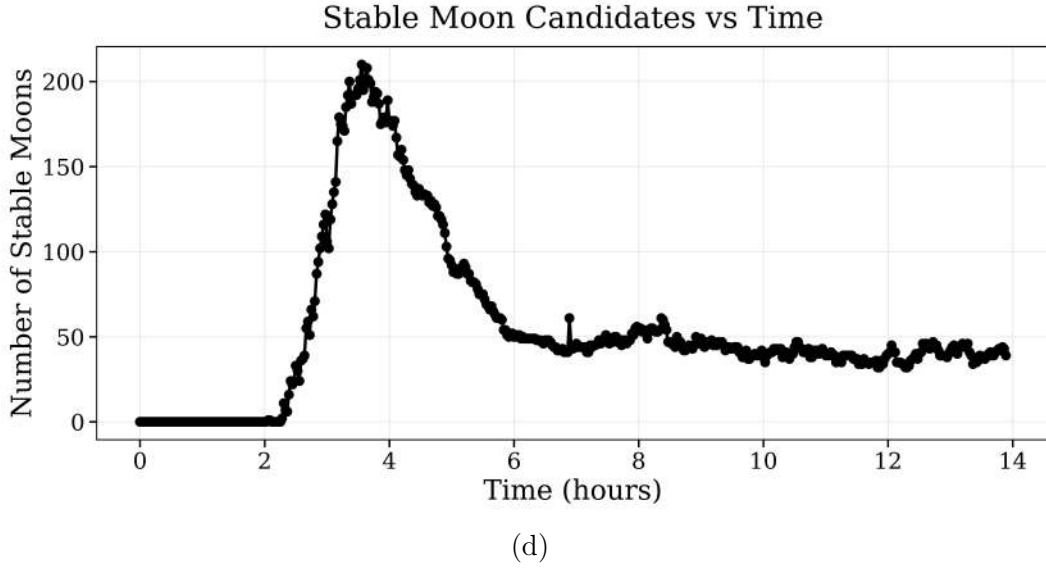
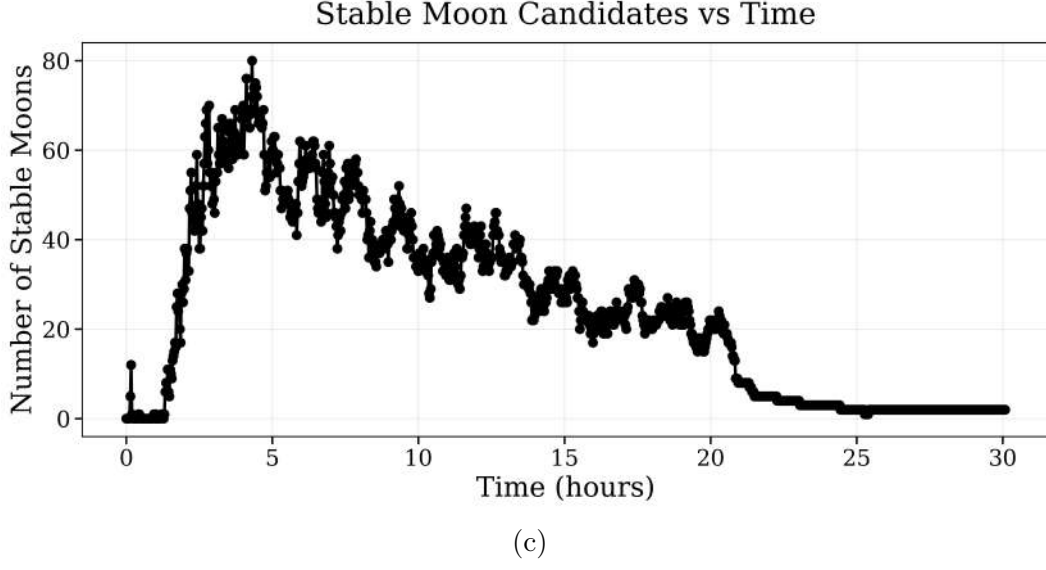


Figure 4: (a), (b), (c), and (d) shows the time evolution of stable exomoon candidates for simulation S2 at grazing angles 15° , 30° , 45° , and 60° , respectively.

4.2.2 Mass Distributions vs. Grazing Angle

Looking at how the final stable masses spread out across the four grazing angles (15° , 30° , 45° and 60°), the specific geometry of a collision shapes the resulting planetary system. In both simulation scenarios, S1 and S2 the primary planet, which is labeled as the “Largest” body stays with a mass of more than 10^{26} kg.

This means that the core of the embryo is strong enough to survive these big impacts without losing much of its mass. However, the story is different for the fragments, which are the potential exomoons. The “Largest” bodies do not grow in a straight line: their development is uneven and highly sensitive to the grazing angle of the collision. The mass of these pieces changes a lot depending on how glancing the blow is. The “Largest” body keeps its mass. The “2nd Largest” bodies are really affected by the collision geometry. In the end the geometry of the collision determines the masses of the planetary system. The “Largest” body and the “2nd Largest” bodies have outcomes. The primary planet stays big and potential exomoons do not grow steadily. So, their growth is affected by the grazing angle of the collision.

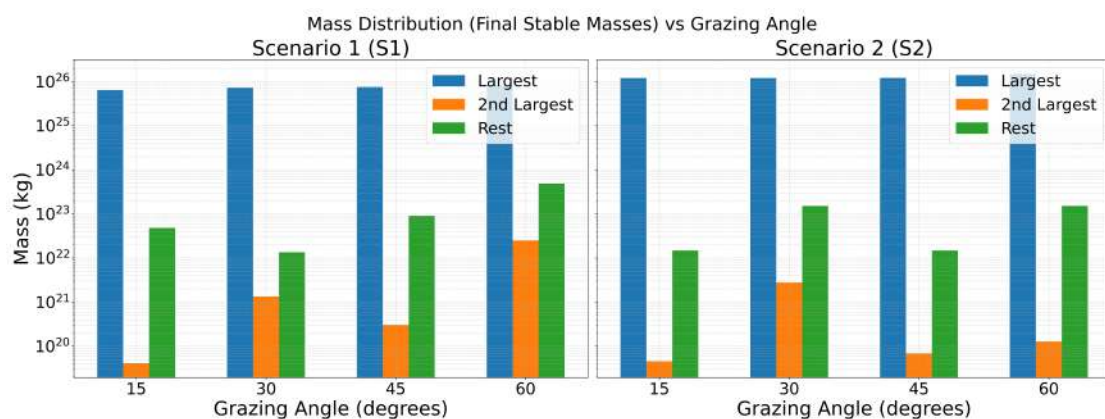


Figure 5: Distribution of final stable masses (Largest, 2nd Largest, and Rest) for Scenarios 1 and 2 across various grazing angles.

Tables 4.1–4.4 show us the mass partitioning between the remnant and the secondary bound object and the remaining debris field for the two collision scenarios S1 and S2. Results are shown for grazing angles of 15°, 30°, 45°, and 60°. The masses I found are compared to things, in our Solar System that have masses so we have something to compare them too.

Table 4.1: Mass distribution for Scenario 1 (S1) and Scenario 2 (S2) at a 15° grazing angle.

Component	S1 mass (kg)	S2 mass (kg)
Primary remnant	6.37×10^{25}	1.19×10^{26}
Secondary	4.13×10^{19} (Mimas)	4.57×10^{19} (Mimas)
Debris field	4.81×10^{22} (Europa)	1.12×10^{22} (Pluto)

Table 4.2: Mass distribution for Scenario 1 (S1) and Scenario 2 (S2) at a 30° grazing angle.

Component	S1 mass (kg)	S2 mass (kg)
Primary remnant	7.27×10^{25}	1.20×10^{26}
Secondary	1.34×10^{21} (Charon)	2.78×10^{21} (Iapetus)
Debris field	1.35×10^{22} (Pluto)	2.83×10^{22} (Triton)

Table 4.3: Mass distribution for Scenario1 (S1) and Scenario2 (S2) at a 45° grazing angle.

Component	S1 mass (kg)	S2 mass (kg)
Primary remnant	7.50×10^{25}	1.21×10^{26}
Secondary	3.07×10^{20} (Vesta)	6.85×10^{19} (Enceladus)
Debris field	9.00×10^{22} (Moon)	1.48×10^{22} (Pluto)

Table 4.4: Mass distribution for Scenario1 (S1) and Scenario2 (S2) at a 60° grazing angle.

Component	S1 mass (kg)	S2 mass (kg)
Primary remnant	8.25×10^{25}	1.47×10^{26}
Secondary	2.50×10^{22} (Triton)	1.27×10^{20} (Enceladus)
Debris field	4.84×10^{23} (Mars)	1.51×10^{23} (Moon)

Something changes in Scenario 1 at a grazing angle of 60°. At this point the biggest fragment is around the mass as 2.50×10^{22} kg, which is roughly the same size, as Neptune’s moon, Triton. This means that steep glancing impacts are the likely to leave behind a large, unified satellite rather than just a cloud of dust. Interestingly there is a dip in mass when hitting at grazing angle the 45°. For example, in Scenario 1 and Scenario 2 the largest body gets smaller from 2.78×10^{21} kg at 30° grazing angle down to just 6.85×10^{19} kg at grazing angle 45°. It seems that at this angle the force of the impact is just right to break the body into tiny fragments instead of letting body stay together as a moon. If we look at the mass of all the smaller fragments, which is the rest category we see a clear trend as wider angles lead to more fragmentation. In Scenario 1 the total mass of the leftover debris at 60° is 4.84×10^{23} kg the leftover debris is almost as massive as Mars. Scenario 2 is a bit calmer with the leftover debris peaking at

1.51×10^{23} kg the mass of the leftover debris is, like the mass of Earth’s Moon. What this tells us is that while both types of collisions, Scenario 1 and Scenario 2 create mess as the angle increases Scenario 1 is far more productive at generating a large family of secondary objects secondary objects like moons. This could lead to systems, with multiple moons or even thick planetary rings.

4.2.3 Trends in Mass Components and Scaling

The Figure 5 provide a useful list of final masses, the logarithmic representation in Figure 6 gives a deeper look into the impact physics. By plotting these components in Earth mass on a log scale, we can identify specific patterns that dictate how these planetary systems evolve after a collision.

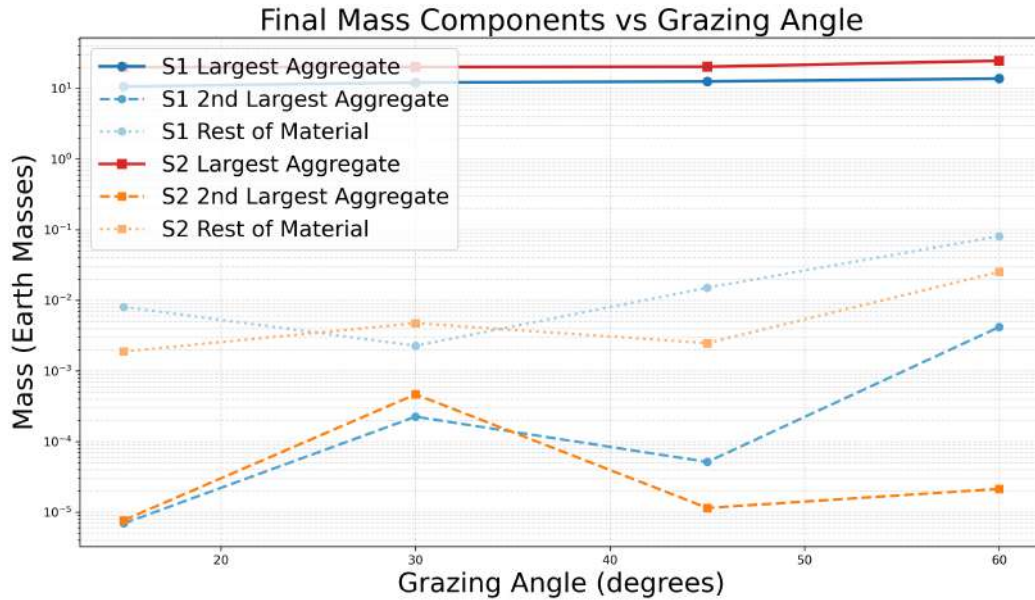


Figure 6: Final mass components for Scenarios 1 and 2 plotted against grazing angle. The logarithmic scale highlights the stability of the primary aggregate relative to the highly variable secondary fragments and debris field.

A key result in both Scenario 1 (solid blue line) and Scenario 2 (solid red line) is the remarkable stability of the primary planet. Whether the impact occurs at grazing angle 15° or grazing angle glancing 60° , the mass of the largest aggregate remains nearly constant. This shows that the planet’s own gravity acts as stabilizing force, letting the growing planet survive even the most violent impacts without losing much of its original mass. In contrast, the “Rest of Material” (dotted lines) represents the most dynamic part of the system. This trend visualizes a

fundamental shift in impact nature at low grazing angle; the process is primarily one of merging, but as we move toward at grazing angle 60° , the collision becomes highly disruptive. At grazing angle 60° the impactor gets broken into fragments creating a huge debris field that is much larger than those from at grazing angle 15° . The information from the biggest fragment (shown with dashed lines) shows that 45° is the critical angle, for destruction. At this specific angle, the collision is too sideways for the two bodies to simply merge, but not sideways enough for the impactor to stay in one piece and form a moon (which often happens at 60°). Instead, the energy from 45° hit is concentrated in just the right way to completely break apart the incoming body into numerous smaller fragments (Canup, 2004).

4.2.4 Particle Distributions for the two Scenarios

The data from the SPH simulations in Figure 7 shows number of particles in Figure 5 and Figure 6 with different grazing angles.

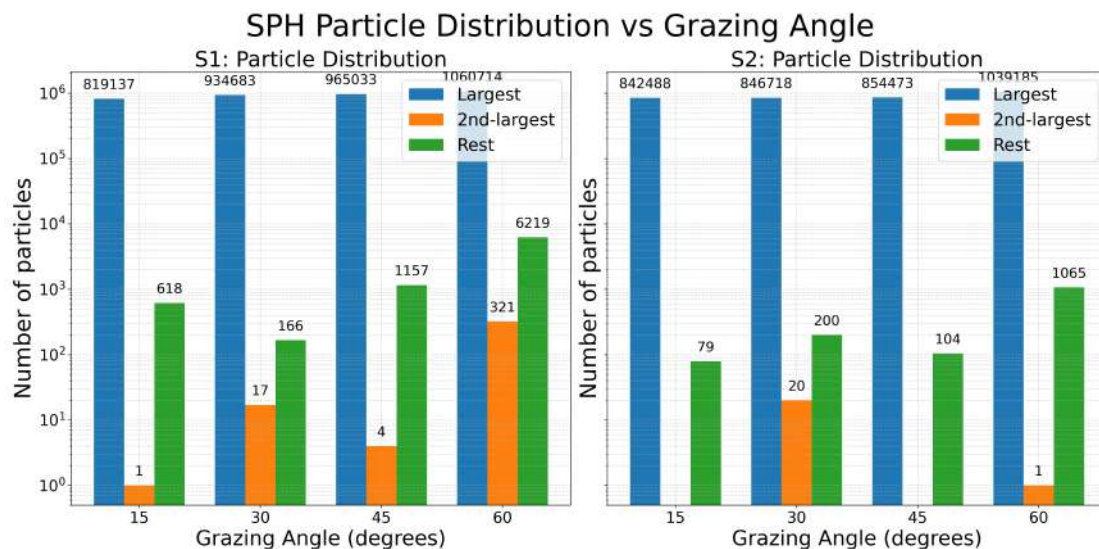


Figure 7: SPH particle distribution for Scenarios 1 and 2 across various grazing angles. The logarithmic scale illustrates the stability of the primary mass (blue) and the significant increase in debris (green) at higher angles.

In both Scenario 1 and Scenario 2, the "Largest Aggregate" remains the dominant feature regardless of the Grazing angle. For Scenario 1 the number of particles in the group stays between 819,137 (when it is at 15°) and 1,060,714 (when it is at 60°). For Scenario 2 the numbers are also steady going from 842,488 to 1,039,185. The angle at which something hits, like the grazing angle affects how much something breaks apart which is noticeable. The level of fragmentation changes when

the steepness of the grazing angle changes and this is true, for Scenario 1 and Scenario 2. At grazing angle 15° Collisions are primarily accretive, resulting in very little debris only 618 particles in Scenario 1 and 79 in Scenario 2. At grazing angle (60°) the impact nature shifts from merging to disruptive. The Rest of Material gets big fast it goes up to 6,219 particles in Scenario 1 and 1,065 particles in Scenario 2. The data shows 45° is an important point where fragmentations start to change for the secondary body. In Scenario 1 the Rest of Material we see that the biggest piece has 17 particles at 30° then it drops down to just 4 particles at 45° but then it jumps, up to 321 particles at 60°). In Scenario 2 the biggest body, the Rest of Material is gone at 45° it has 0 particles.

Table 4.5: SPH Particle Counts per Simulation Aggregate

Simulation	Largest Aggregate	2nd-Largest Aggregate	Rest of Material
S1_15	819,137	1	618
S1_30	934,683	17	166
S1_45	965,033	4	1,157
S1_60	1,060,714	321	6,219
S2_15	842,488	0	79
S2_30	846,718	20	200
S2_45	854,473	0	104
S2_60	1,039,185	1	1,065

4.2.5 Disk Mass vs. Grazing Angle

In this section, we can examine how the steepness of the impact the grazing angle affects the total mass of the resulting debris disk. Looking at the data for both Scenario 1 ($12 M_\oplus$) and Scenario 2 ($20 M_\oplus$), the mass changes a lot depending on the angle.

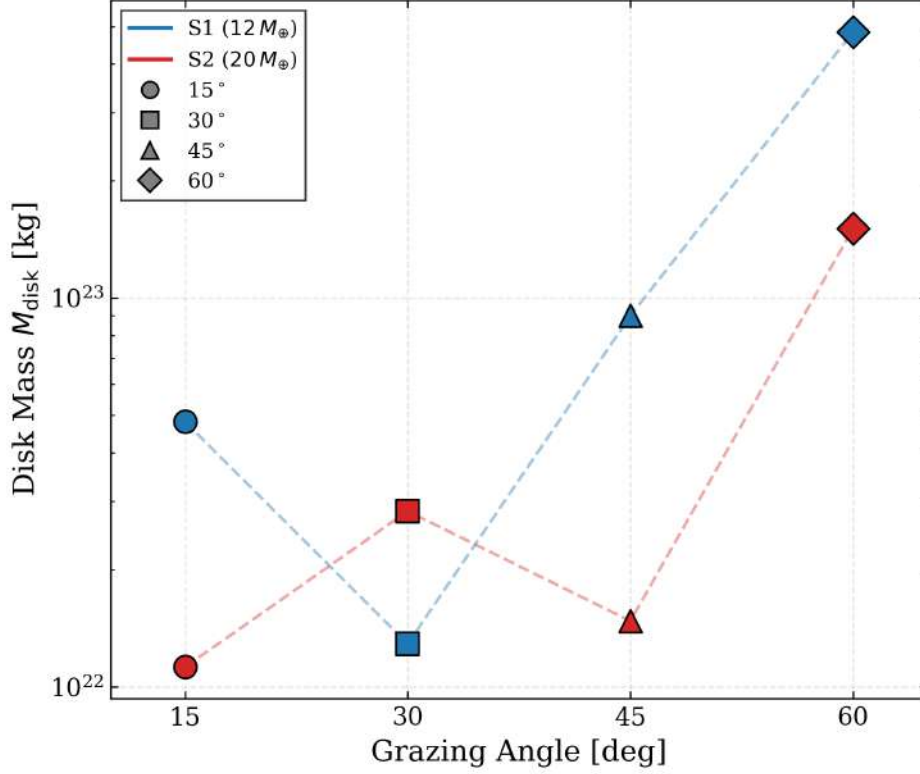


Figure 8: Total disk mass (M_{disk}) in kilograms plotted against the grazing angle (degrees) for scenarios Scenario 1 ($12M_{\oplus}$) and Scenario 2 ($20M_{\oplus}$).

For both scenarios, the highest disk mass is recorded at the steepest grazing angle of 60° . This suggests that more "glancing" blows at higher angles are more efficient at stripping material away into a stable orbit rather than having it fall back onto the planet. The two scenarios do not behave the way in the middle range. Scenario 1 has a drop in mass at 30° and then it goes up slowly, but Scenario 1 does this in a steady way. On the hand Scenario 2 has a small peak at 30° and then it goes down at 45° . Even though Scenario 2 starts with a mass, which is $20 M_{\oplus}$ it does not always make a heavier disk than Scenario 1 and Scenario 2. For instance, at 15° , the Scenario 1 is results in a more massive disk than the heavier Scenario 2.

4.2.6 Particle Count vs. Grazing Angle

While disk mass tells us the "weight" of the debris, the particle count gives us a better look at the distribution and density of that material, specifically whether it stays near the planet or escapes beyond the Roche limit (R_r).

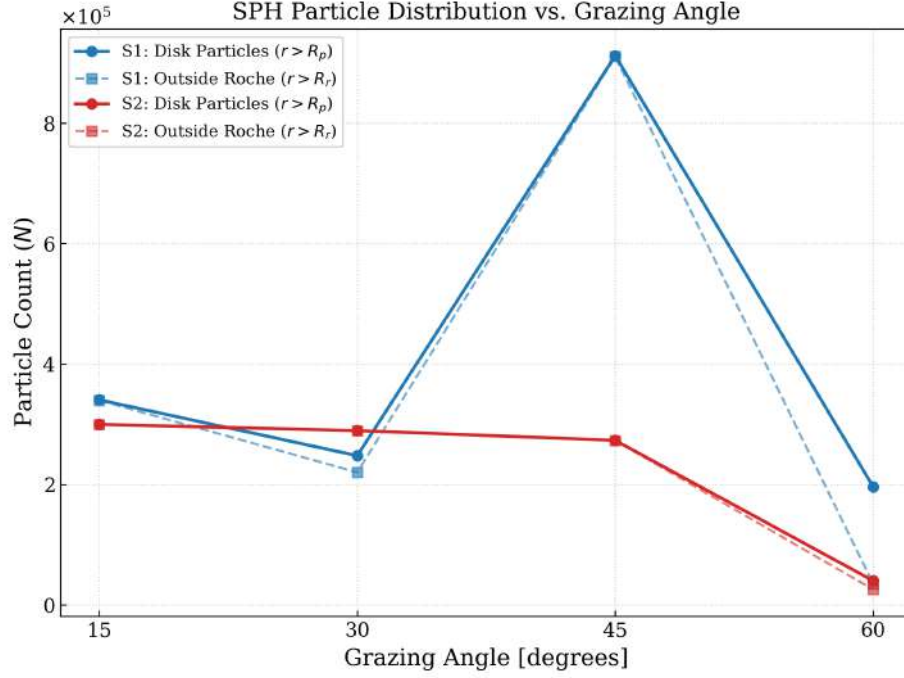


Figure 9: Distribution of SPH particles (N) vs. grazing angle. The plot compares total disk particles ($r > R_p$) and those located outside the Roche limit ($r > R_r$) for both S1 and S2 scenarios.

Table 4.6: Disk and Roche-exterior particle count for different Grazing angles

Scenario	Grazing Angle ($^\circ$)	Disk Particles ($r > R_p$)	Particles outside Roche ($r > R_r$)
S1 (Blue)	15	$\approx 3.4 \times 10^5$	$\approx 3.4 \times 10^5$
	30	$\approx 2.5 \times 10^5$	$\approx 2.2 \times 10^5$
	45	$\approx 9.1 \times 10^5$	$\approx 9.1 \times 10^5$
	60	$\approx 2.0 \times 10^5$	$\approx 0.4 \times 10^5$
S2 (Red)	15	$\approx 3.0 \times 10^5$	$\approx 3.0 \times 10^5$
	30	$\approx 2.9 \times 10^5$	$\approx 2.9 \times 10^5$
	45	$\approx 2.7 \times 10^5$	$\approx 2.7 \times 10^5$
	60	$\approx 0.4 \times 10^5$	$\approx 0.2 \times 10^5$

The particle data is the peak for Scenario 1 at grazing angle 45° with about 9.1×10^5 particles. At this angle all the particles from the disk are outside the Roche limit resulting in a big and spread-out debris field. The particle counts for Scenario 2 which is the red line are much more steady at 15° , 30° , and 45°

ranging from 2.7×10^5 to 3.0×10^5 . Scenario 2 does not have the ups and downs that Scenario 1 has but it does drop sharply when the angle gets to 60° . In fact, at grazing angle 60° the particle count is low for both Scenario 1 and Scenario 2. This means that at 60° the disk is made up of particles or pieces rather than a lot of small debris. At angle 15° , almost all the particles made stay outside the Roche limit for both Scenario 1 and Scenario 2. This tells us that the energy, from the impact is enough to push the material away, so it does not get absorbed right back in.

4.3 Deeper Analysis: Evolution of the Debris Disk

The evolution of the debris disk can be understood as a gradual change from an organized structure into a more chaotic and dispersed system. By examining how the mass density changes with respect to both the Roche limit and the planet's radius, we can identify when tidal forces become strong enough to break apart the secondary body.

4.3.1 Early Phase: Structural Integrity

At the beginning of the simulation, the system remains well structured. The mass fraction plots show narrow and clearly defined peaks. Most of the mass is located either very close to the planet's surface or safely within the Roche limit, in the range $R/R_{\text{Roche}} \approx 0.25\text{--}1.45$ (Table 4.7).

The regions between these peaks contain very little material, which indicates that the objects are still separate and stable. The numerical results support this observation. For example, simulation S1 at grazing angle 15° begins with a disk mass of 1.7310×10^{25} kg and a particle count of $N_{\text{disk}} = 348,519$ (Table 4.8).

At this stage, the secondary body is still intact and maintains its structural integrity, with only minimal spreading of material.

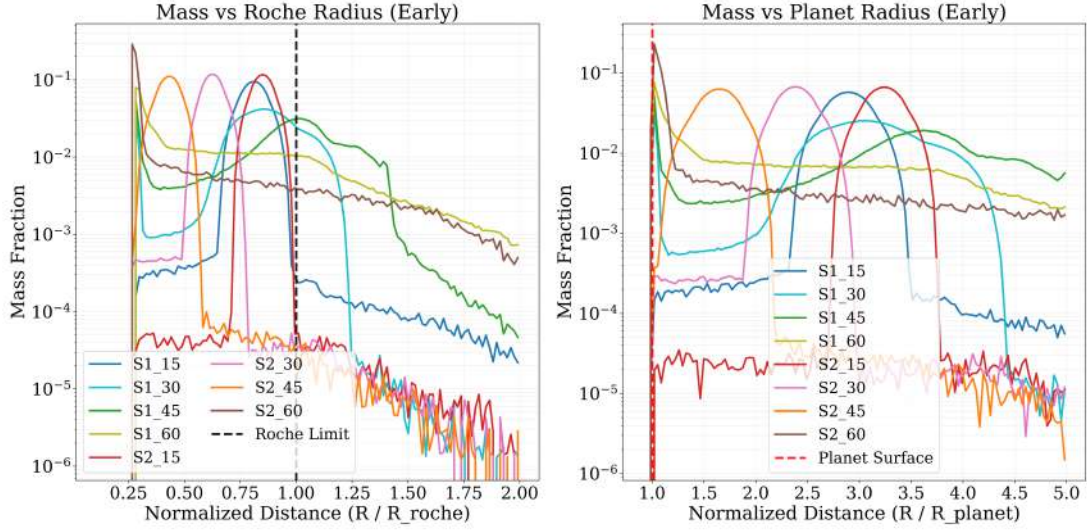


Figure 10: Mass fraction at the Early stage, showing distinct peaks.

4.3.2 Middle Phase: Beginning of Tidal Disruption

As time progresses, the system begins to lose its original structure. The sharp mass peaks become wider and less distinct. The mass fraction decreases in some regions to values as low as 10^{-6} , while the material extends farther outward, reaching distances up to $4.80 R_{\text{planet}}$ (Table 4.7).

This behavior indicates that tidal forces from the planet are starting to remove material from the outer layers of the secondary body. This process forms a tidal stream of debris. The simulation data shows a clear increase in particle number. For instance, simulation S1 at grazing angle 30° reaches $N_{\text{disk}} = 475,142$ (Table 4.8), which reflects the breaking apart of the original solid body into many smaller fragments.

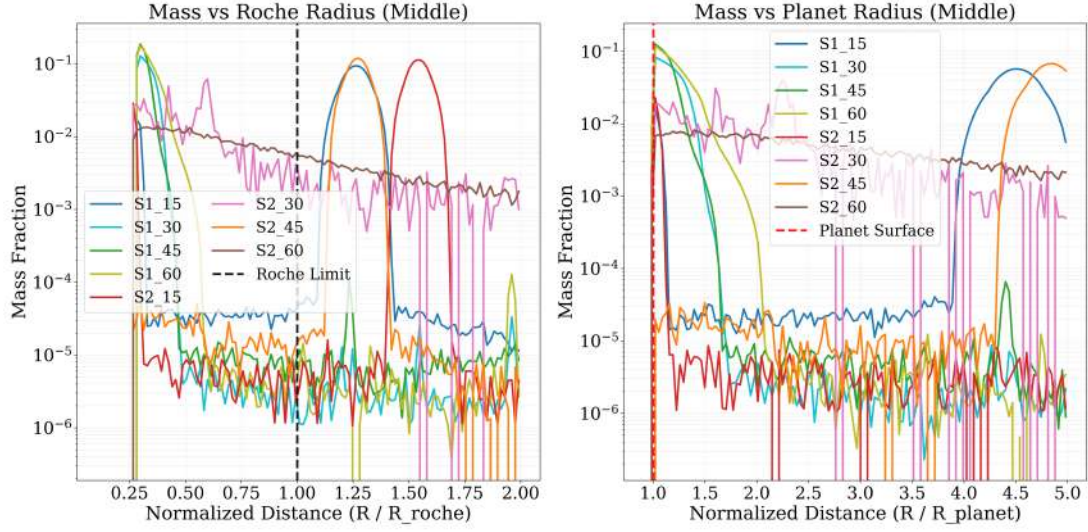


Figure 11: Mass fraction at the Middle stage, highlighting the onset of tidal disruption and the rising density floor between reservoirs.

4.3.3 Late Phase: Final Dispersal and Accretion

In the final phase, the original body is no longer present as a single object. Instead, the material is fully dispersed throughout the orbital region, forming a low-density debris disk that extends out to approximately $5.0 R_{\text{planet}}$ (Table 4.7).

Two important processes dominate this stage:

1. **Fragmentation** The particle counts increases dramatically, indicating complete breakup. For example, simulation S2 at grazing angle 15° reaches $N_{\text{disk}} = 1,053,120$ (Table 4.8), which confirms extensive fragmentation.
2. **Accretion** Accretion happens when some of the material moves inward toward the planet, at $R/R_{\text{planet}} = 1.0$ (Table 4.7). This means that some of the debris falls onto the surface of the planet or the planet absorbs it.

The simulation really shows us what happens when a stable body turns into a debris disk that is completely dispersed because of forces. The Accretion process is happening and changes everything.

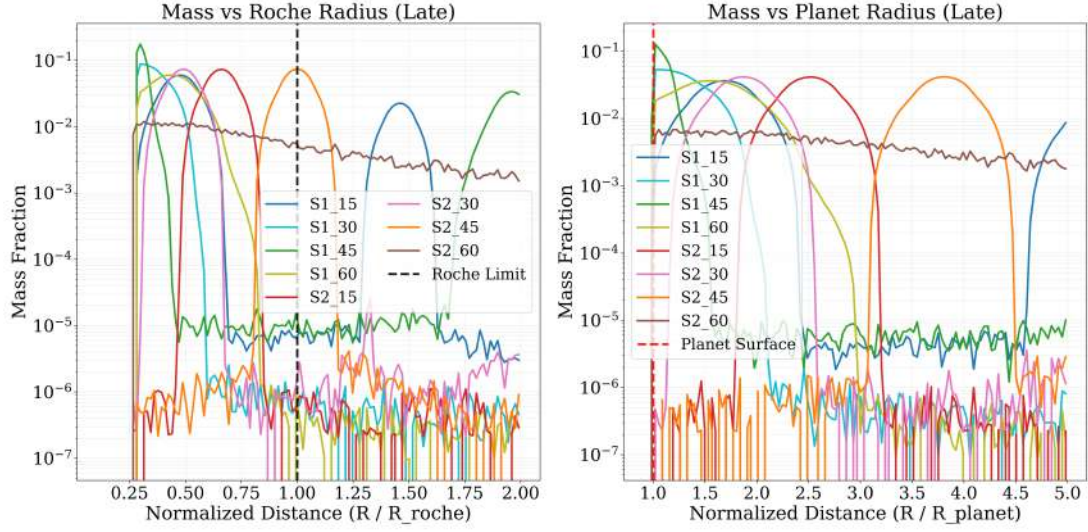


Figure 12: Mass fraction at the Late stage, demonstrating a fully formed debris tail and mass accumulation at the planetary surface.

Table 4.7: Evolution of Peak Mass Fraction and Spatial Distribution throughout the simulation phases.

Stage	Peak Fraction ($\approx$)	Mass ($\approx$)	Roche Range (R/R_{Roche})	Planet Range (R/R_{planet})	Observations
Early	10^{-5} – 2×10^{-1}		0.25–1.45	1.00–4.25	most mass near planet surface or within Roche limit.
Middle	10^{-6} – 10^{-1}		0.25–1.60	1.00–4.80	peaks flatten and shift outward
Late	10^{-7} – 10^{-1}		0.25–1.90	1.00–5.00	material widely spread

4.3.4 Quantitative Evolution of the Debris Disk

The mass density plots for each simulation in Table 4.8. While the Figures 10–12 show where the debris is. Table 4.8 tells us how debris there is and how broken up the debris is at each stage and looks at three numbers for every simulation. The first number is the mass of the debris disk (M_{disk}), the second number (N_{disk}) is how many pieces of debris are in the debris disk, and third number (N_{Roche}) is how many of these pieces of debris are near the planet and the Roche lobe which's a dangerous area called the tidal tearing zone. By looking at these numbers from

the start, to the end we can see exactly what happens when the debris is torn apart by the planet's gravity.

Table 4.8: Disk Properties per Simulation Stage

Simulation	Stage	Disk Mass (kg)	N_{disk}	N_{Roche}	N_{total}
S1_15	Early	1.731×10^{25}	348,519	5,605	1,075,462
S1_30	Early	7.0833×10^{24}	246,323	61,130	1,075,637
S1_45	Early	5.0671×10^{24}	240,943	118,950	1,075,637
S1_60	Early	3.9596×10^{24}	255,003	85,563	1,075,637
S2_15	Early	2.3503×10^{25}	301,154	754	1,053,139
S2_30	Early	2.3027×10^{25}	293,402	583	1,053,139
S2_45	Early	2.174×10^{25}	274,375	401	1,053,139
S2_60	Early	2.4138×10^{24}	51,064	9,502	1,053,139
S1_15	Middle	1.7602×10^{25}	362,061	339,936	1,075,462
S1_30	Middle	1.9609×10^{25}	475,142	220,405	1,075,637
S1_45	Middle	1.2575×10^{25}	377,480	172,307	1,075,637
S1_60	Middle	1.684×10^{25}	402,141	38,540	1,075,637
S2_15	Middle	2.4001×10^{25}	319,602	299,934	1,053,139
S2_30	Middle	5.3169×10^{22}	1,533	616	1,053,139
S2_45	Middle	2.1539×10^{25}	273,589	272,992	1,053,139
S2_60	Middle	2.3093×10^{24}	42,387	25,448	1,053,139
S1_15	Late	7.1575×10^{25}	1,072,129	339,758	1,075,462
S1_30	Late	4.232×10^{25}	714,777	220,266	1,075,637
S1_45	Late	9.3135×10^{24}	326,625	171,790	1,075,637
S1_60	Late	6.522×10^{25}	920,949	35,081	1,075,637
S2_15	Late	1.1944×10^{26}	1,053,120	299,953	1,053,139
S2_30	Late	1.1944×10^{26}	1,053,097	289,472	1,053,139
S2_45	Late	1.1944×10^{26}	1,053,133	651,286	1,053,139
S2_60	Late	2.1885×10^{24}	40,163	25,601	1,053,139

Table 4.8 gives us a picture of how the system changes. It starts with a body and ends up as a big cloud of debris. At the beginning the damage is not too bad. For example, in run S1_15 there are about 348,519 fragments of debris and is one third of the total material. The disk is not very heavy it has a mass of 1.731×10^{25} . This means the secondary body is still mostly in one fragment it is just starting to break apart a little. As the simulation moves on to the part, we can see that things start to fall apart. If we look at grazing angle 30° , which is called S1 the number of fragments of debris goes up to 475,142. What is interesting is that the number of particles that are trapped near the planet goes up to 220,405. This is an increase and it shows that the planets gravity is pulling hard on the

secondary body. This is what we call stripping. It is making a mess and spreading the debris all over the place. By the end Table 4.8 shows us two things that can happen to the system. In some cases, like S2 at grazing angle 15° the body breaks apart completely. There are 1,053,120 fragments of debris which is almost all the original material. This means the body has been totally broken into pieces. But in cases like S1 at grazing angle 60° things are different. The number of fragments of debris is lower, at 920,949. There is a big group of 35,081 particles that are still stuck together. This means that the body did not break apart completely. The outside layers were stripped away. The core was strong enough to survive.

4.4 Analysis of Disk Vapor Fraction Stability

Figure 13 tracks the vapor fraction over time across eight simulations, which are grouped into S1 and S2 over a total timeframe of roughly 42 hours. At first the system stays entirely in the vapor phase for 8.44 hours. This shows up as a yellow block meaning the vapor fraction is almost 1.0. The system starts in an energetic state, and we do not see much change regardless of the starting conditions. Then between 8.44 and 29.55 hours we can see vertical bands. These bands mean the vapor fraction is changing rapidly across the disk. Because we see these patterns at the time in all our scenarios the changes are likely due to the passing of time in the system not the unique variables of each run.

After 29.55 hours the runs start behaving. The stable setups, like S1 and S2 at grazing angle 30° keep a high vapor fraction meaning the material stays mostly gaseous. In other runs like S2 at grazing angle 60° and S1 at grazing angle 15° the vapor fraction drops shifting to a light green color on the plot. This drop means the material may be starting to condense in areas or the system is losing its thermal efficiency under those conditions. The Roche limit and the secondary body are still playing a role, in the disk vapor fraction stability. The planets gravity and the tidal stripping are affecting the vapor fraction and the debris field. The secondary body and the disk fragments are still interacting with each other.

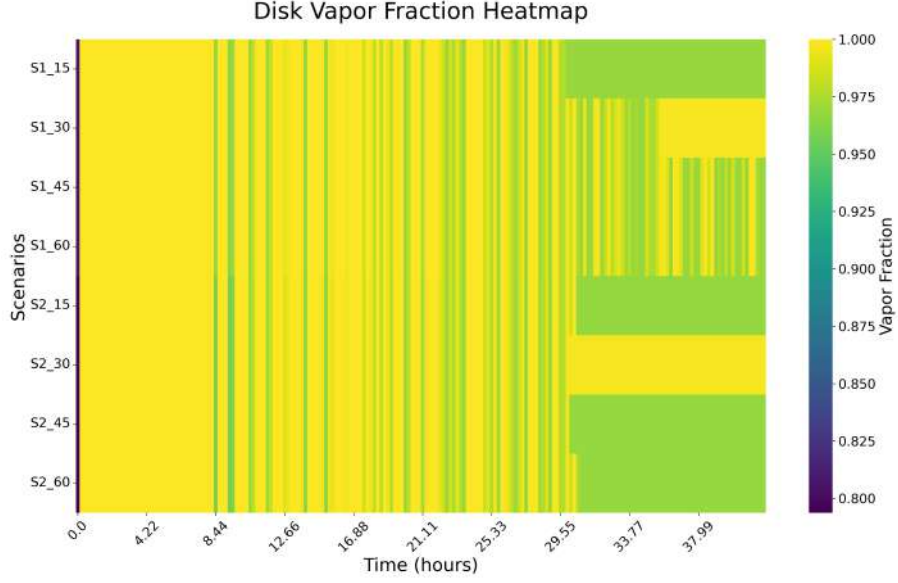


Figure 13: Heatmap illustrating the temporal distribution of disk vapor fractions across Scenarios (S1 and S2).

The behavioral stages of the system are further synthesized in Table 4.9, which categorizes the timeline into three distinct operational windows based on the observed stability and divergence of the vapor fraction.

Table 4.9: Phase Classification of Vapor Fraction Stability

Phase	Time Range (h)	Observation
Saturation	0.0 – 8.44	Uniform vapor fraction maintained across all eight simulations.
Instability	8.44 – 29.55	Identification of periodic fluctuations.
Divergence	29.55 – End	Scenarios diverge into distinct high vapor and reduced vapor conditions.

4.5 A Comparative Analysis of Fragment Stability and Orbital Architecture in Tidal Disruption Events: S1 vs. S2 Simulation Scenarios

The results from the Scenarios 1 and 2 gives the details how fragments of a celestial body are distributed and categorized after a tidal disruption event. These results

are explained through three main lenses: fragment stability, orbital characteristics, and mass distribution.

4.5.1 Stability Classification

The primary outcome of each simulation is the classification of fragments based on their long-term orbital outlook:

1. Stable Fragments: Fragments that remain in a predictable, permanent orbit around the central planet.
2. Marginal Fragments: Fragments in a “transition” state, possessing orbits that are neither clearly safe nor immediate failures.
3. Unstable Fragments: Objects destined to either collide with the planet or be ejected from the system entirely.

4.5.2 Orbital Architecture and Parameters

The Orbital Architecture plots provide a visual map of where these fragments reside relative to the planet’s gravitational “danger zones”:

1. Roche Limits: The reports reference the Fluid and Rigid Roche Limits. These represent the physical boundaries where the planet’s tidal forces become strong enough to overcome the fragment’s internal gravity. Objects found within these limits (indicated by the dashed and dotted circles) are highly break up or unstable.
2. Semi major Axis (a) and Eccentricity (e): These parameters describe the “size” and “stretch” of the new orbits.

For Scenario 1: Characterized by high-eccentricity disruptions. For example, at grazing angle 30° , the surviving object reaches a semi-major axis of approximately $50R_p$ with a very high eccentricity between 0.88 and 0.96.

For Scenario 2: Explores more circular fragment distributions. At grazing angle 30° , fragments eccentricity between 0.1 and 0.7, indicating more stable, rounded paths.

4.5.3 Simulation Scenario Analysis Scenarios 1 and 2

The following figures provide a comprehensive breakdown of the two distinct scenarios. The Figure 14 and Figure 15 illustrates that the fate of "massive fragments"

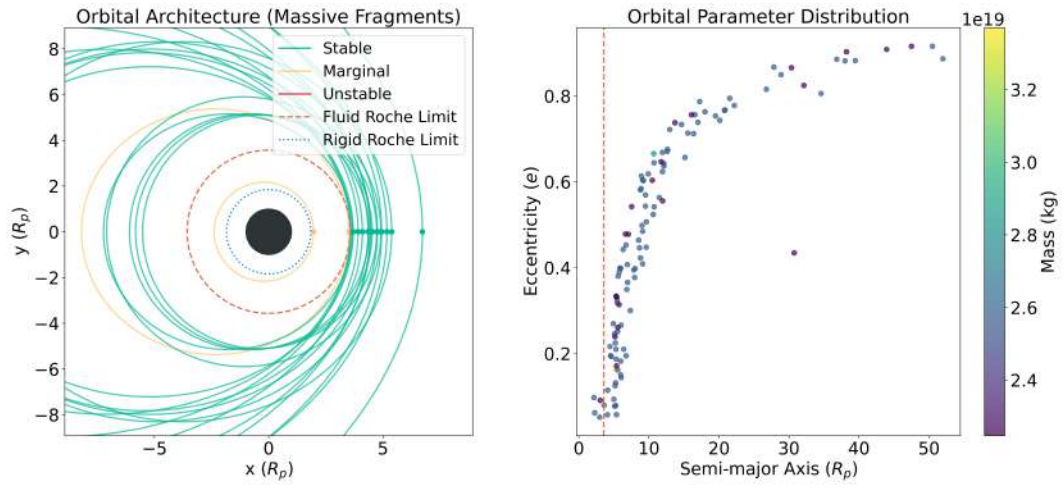
resulting from a celestial collision at four different grazing angles (15° , 30° , 45° , and 60°). They specifically track which debris remains in a stable orbit, which is marginally stable, and which is likely to be destroyed by the planet's gravity.

Evolution of Debris Architecture across Grazing Angles For Scenario 1:

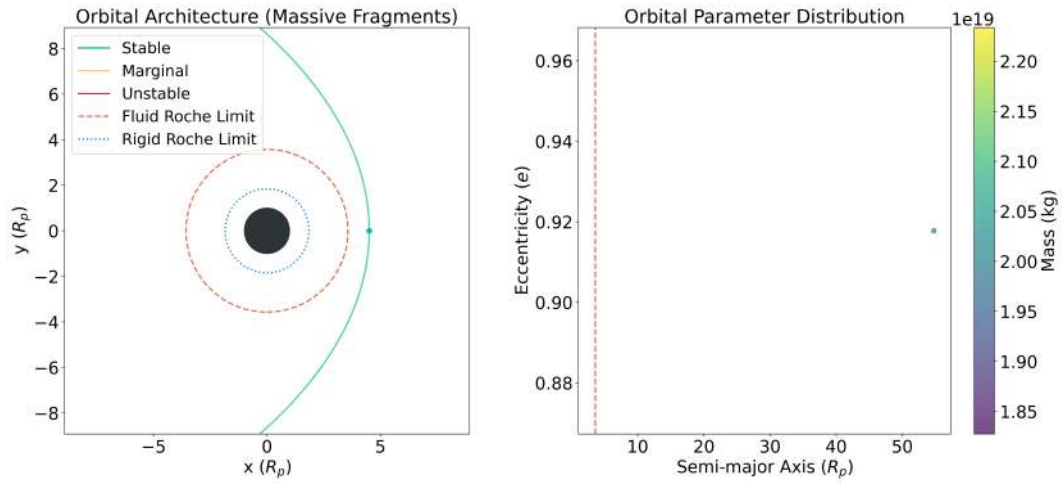
At 15° grazing angle, the collision produces a massive number of debris being scattered across a wide range of distances. The orbital architecture plot shows a dense web of orbits, many of which extend far beyond the planet. Because the impact happens at low grazing angle, the fragments move with many different speeds and in different directions, creating a distribution where eccentricity increases sharply with the distance from the planet. We can see lots of fragments of something sitting on the edge of the "Fluid Roche Limit". This tells us that while many of these fragments survive a lot of them are being pulled hard by tidal forces. As the angle gets bigger to 30° the whole system starts to look more organized and less chaotic. The simulation shows that far fewer large fragments remain after the collision. At this steeper angle, the impact energy is more focused, so fewer separate groups of debris are formed. The one remaining large fragment moves in a very stretched, elliptical orbit far from the planet. It is also safely outside the Roche limit, where the planet's gravity could break it apart. This tells us that when something hits a planet hard it usually leaves just one piece of debris far away from the planet instead of making a lot of pieces that form a dense ring around the planet.

At an angle of 45° there are fragments of debris more than when it hits at 30° . These fragments of debris stay close to each other. They orbit the planet at about the same distance. The paths they follow around the planet are not as stretched out and are, like a circle, which makes them more stable. This shows that hitting a planet at 45° may be the case, where the debris is spread out just enough to form a small and stable group of objects that orbit the planet.

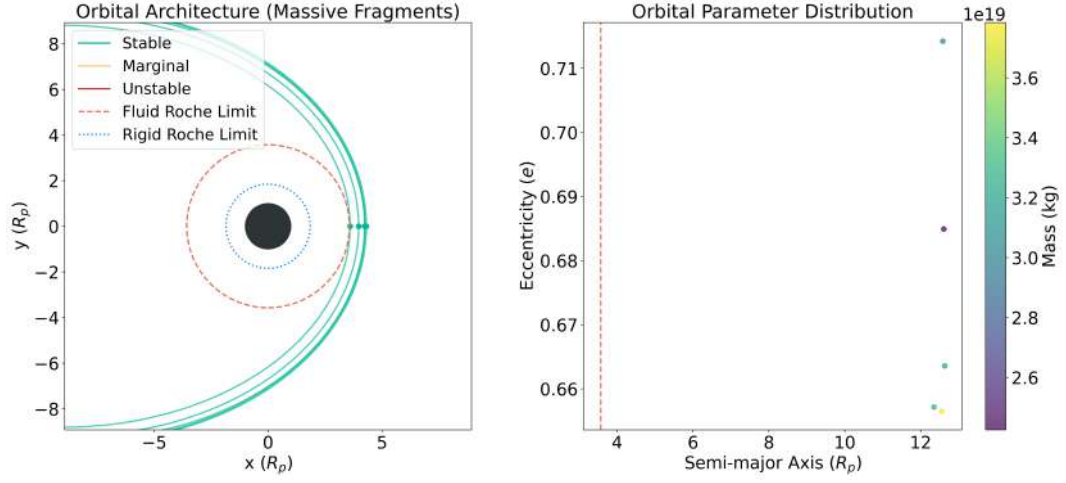
Finally, at 60° suggests that steeper impacts usually leave behind one distant fragment instead of many fragments forming a dense ring around the planet. The results show only a couple of massive fragments surviving, but they are notably more massive than those in the lower angles, as shown by the yellow-to-purple color scale. These fragments have moved further out in distance but maintain high stability. The increased angle seems to filter out smaller debris, leaving behind only the most substantial fragments in well-defined, distant orbits.



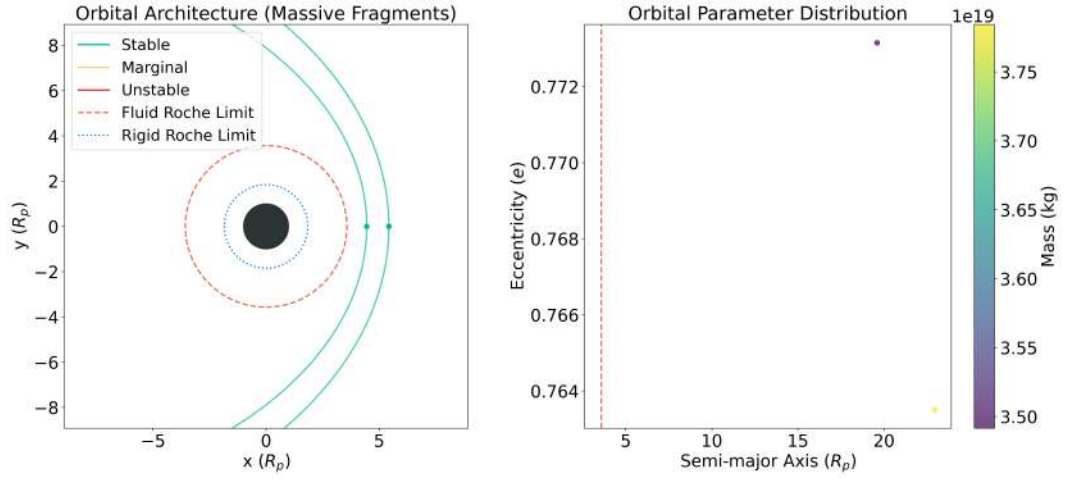
(a)



(b)



(c)



(d)

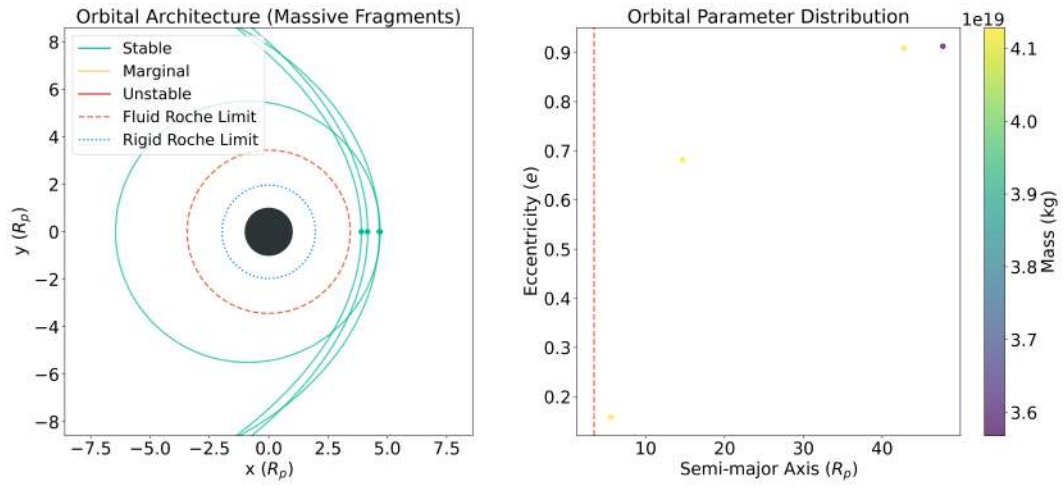
Figure 14: (a), (b), (c), and (d) show the fragmentation and orbital architecture of massive fragments for grazing angles of 15° , 30° , 45° , and 60° for Scenario1 respectively.

Evolution of Debris Architecture across Grazing Angles For Scenario

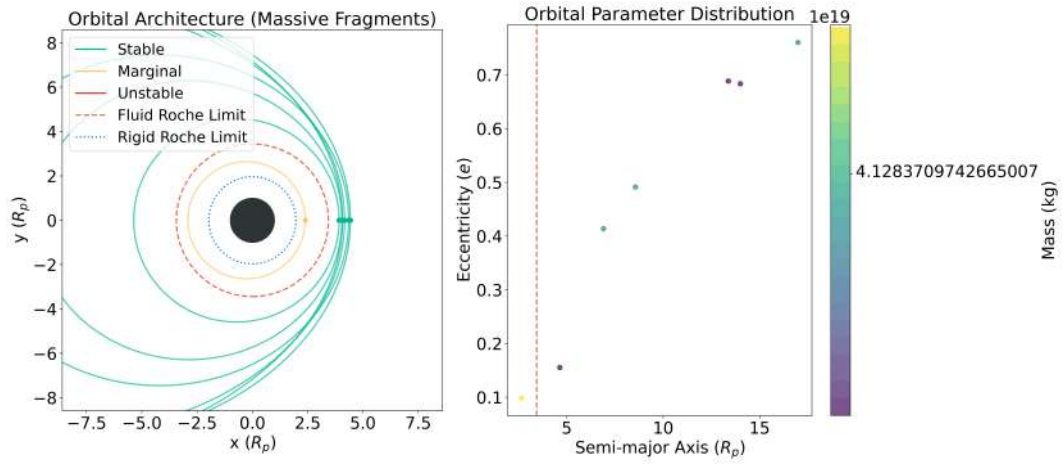
2:

At 15° grazing angle, only a few large and stable fragments are formed. These fragments are away from each other with one fragment close to the planet and the other fragments very far away almost 50 times the distance from the planet to its edge. This big distance between the fragments shows that they have a lot of energy to move far away, from each other and stay in orbits that do not change much. Because they are far apart the fragments do not affect the fragments much and the fragments can stay stable for a long time. Increasing the angle at 30° , the fragments are set up a lot by making it so, the fragments are just marginal stability. While most fragments remain stable and green on the plot, one fragment is positioned dangerously close to the Fluid Roche Limit, as shown by the yellow orbit. The parameter distribution reveals a clear trend where fragments further from the planet have higher eccentricity. The 30° impact makes an area of debris where the pieces of debris are more likely to run into each other because of gravity. When the impact is at 45° the debris goes over the place, and the orbits are not neat. The fragments are over the place between 40 and 70 planetary radii. This means that when the impact is at 45° in Scenario 2 it is strong enough to throw the fragments away in many directions. Even though the orbits are very different they all stay outside the area, around the planet where the strong tidal forces could hurt them.

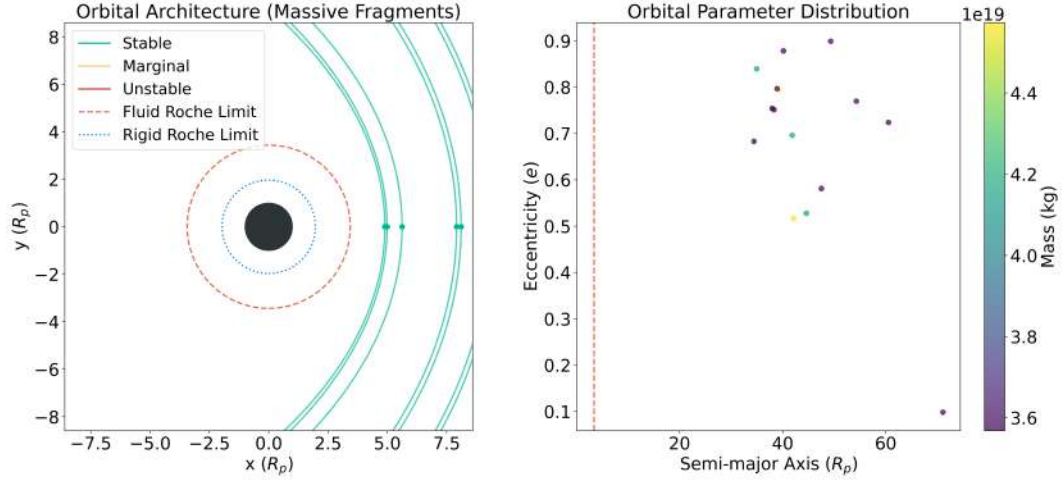
The 60° impact is different. It makes number of debris, so the orbits are very crowded and complicated. The architecture plot is filled with orange “marginal” orbits, indicating a high density of material packed tightly around the planet’s Roche limits. The plots of how things are spread out shows that there are a lot of fragments that are heavy like they weigh around 10^{20} kg and they are all close together near the parent body. This means that at an angle of 60° it makes a lot of debris that stays near the parent body. This could be how a ring or disk system that lasts a time is formed. The debris from the collision stays close, to the parent body, which is what we see in the distribution plot of the fragments.



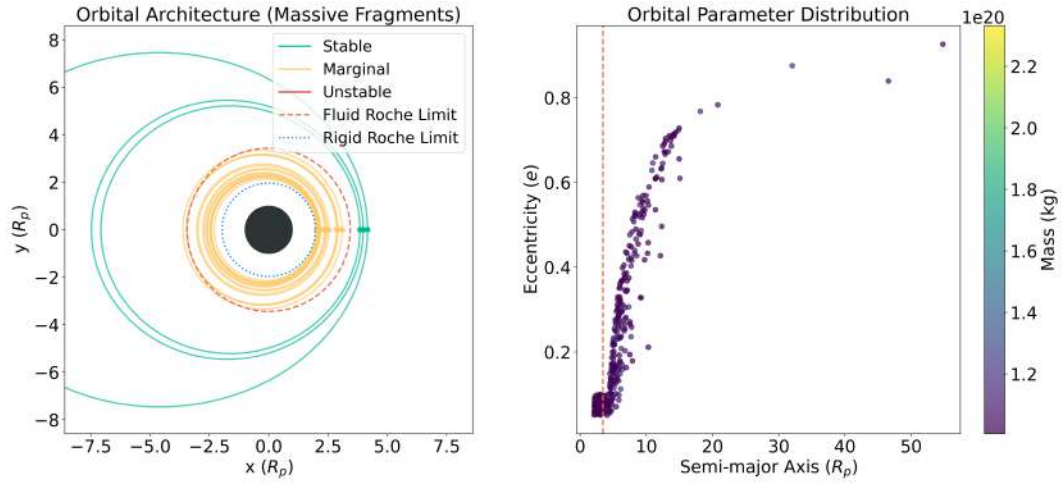
(a)



(b)



(c)



(d)

Figure 15: (a), (b), (c), and (d) show the fragmentation and orbital architecture of massive fragments for grazing angles of 15° , 30° , 45° , and 60° for Scenario2 respectively.

Table 4.10: Dynamical Outcomes and Mass Distribution of Tidal Disruption Simulations

Simulation Case	Fragment Count	Primary Dynamical Fate	Largest Mass Range (kg)
S1_15	~ 120	Stable Orbit	$2.4 \times 10^{19} - 3.4 \times 10^{19}$
S1_30	1	Orbital Consolidation	2.03×10^{19}
S1_45	~ 5	Stable Chunks	$2.4 \times 10^{19} - 3.8 \times 10^{19}$
S1_60	~ 2	Stable Binary/Pairs	$3.5 \times 10^{19} - 3.75 \times 10^{19}$
S2_15	~ 4	Stable Orbit	$3.6 \times 10^{19} - 4.1 \times 10^{19}$
S2_30	~ 7	Stable Orbit	4.128×10^{19}
S2_45	~ 14	Stable Orbit	$3.6 \times 10^{19} - 4.6 \times 10^{19}$
S2_60	~ 39	Stable Orbit	$1.0 \times 10^{20} - 2.4 \times 10^{20}$

When something like a moon goes past the Roche limit that is when fragmentation starts to happen. The planets gravity is strong at this point. It is stronger, than the gravity of the object itself. In Scenario 1, the simulations tend to keep large, intact fragments. In Scenario 2, the destruction is more severe, at grazing angle 60° case being extreme breaking the original body into lots of fragments. Even after breaking apart, the largest fragments in most cases stay gravitationally stable. This means they could form a long-lasting debris structure or even a ring around the planet.

Chapter 5

Discussion

5.1 Interpretation of Results

5.1.1 Geometry of Impact and the Partitioning of Mass

The simulation results indicate a deep relationship between the initial collision geometry and the resulting distribution of mass within the system. Specifically, the grazing angle acts as a primary lever for satellite formation. At grazing angle 15° , both Scenarios 1 and 2 (S1 and S2) failed to produce significant orbital material, yielding secondary bodies roughly equivalent to Saturn’s Mimas (4×10^{19} kg) in Table 4.1. These “direct” hits appear to consolidate most of the mass into the primary planet, leaving little for moon formation. When the collision is not on the result is different. At an angle of 60° in S1 the field of debris gets a lot bigger. It gets as big as 4.84×10^{23} kg which is shown in Table 4.4. This is the same as the mass of Mars. This shows us that what I learned from (Barr and Syal, 2017) is true. I see that when things hit at a glancing angle it is much better at keeping material floating in space. The collision and the debris field and the material in orbit, like Mars are all affected by this. By avoiding immediate re-absorption by the primary body, these high-angle collisions create the “storage” of mass necessary for substantial moon growth.

5.1.2 Evaluating the Potential for Exomoon Formation

A surprising thing I found in the data is that S1, the Optimized Super-Earth that’s $12 M_\oplus$ does a better job than S2, the Giant Impactor Limit that is $20 M_\oplus$ at making debris at high angles. Even though S1 has total mass in the system it made a debris field at a grazing angle of 60° that was about three times more massive than what S2 made. The debris field from S1 is big like the size of Mars. It also made a moon that is like Triton in size. On the hand S2, the biggest moon is only as big,

as Enceladus, which is much smaller. This suggests a gravitational “sweet spot” for moon formation. In S2, the primary remnant’s immense gravity likely pulls material back toward the surface before it can stabilize in orbit. These supports the hypothesis by (Malamud et al., 2020) that larger Super-Earths face greater hurdles in moon formation, requiring extremely specific, high-angular-momentum events to produce satellites of any significant size.

5.1.3 Consequences for Observational Astronomy

The distinction between these two scenarios is critical when considering current detection limits. Most modern transit techniques require a moon to be at least 0.1 Earth mass to be identified with confidence.

Scenario 1: At grazing angle 60° , the debris field (4.84×10^{23} kg) in Table 4.4, nears this detection threshold, suggesting that the resulting moon could eventually be observable.

Scenario 2: Even at its optimal grazing angle 30° , the debris field (2.83×10^{22} kg) in Table 4.2, remains an order of magnitude too small for current instruments. This means that smaller Super-Earths could be the Super-Earth to look for exomoons. This is an idea because people might think that bigger Super-Earth is better for finding exomoons. This new idea helps us make our search for exomoons. It tells us that we should look at Super-Earth where the stars gravity is not too strong. This allows big clouds of debris to form around the Super-Earth (Kipping et al., 2012). So, we should look for exomoons, around Super-Earth not just look for Super-Earth.

5.1.4 Long-Term Stability and Satellite Evolution

The secondary objects identified in my results, such as the Charon-mass moon in S1 at 30° , represent only the beginning of the satellite’s life cycle. For these bodies to stay around they must deal with the conditions that happen right after they crash into something. The rules that (Heller and Barnes, 2013) and (Sasaki and Barnes, 2014) mention state that a moon can only exist for a period if it is in the right position relative to the Roche limit. The Roche limit is the distance within which a planets forces would tear a moon apart. So, the long-term survival of a moon like this moon is decided by where it’s in relation, to the Roche limit.

5.1.4.1 The Multi-Clump Pathway

My simulations show that high grazing angle (45° – 60°) produce expansive, mass-rich debris fields in Table 4.3 and Table 4.4. In these scenarios, it is unlikely that

a single, large moon forms instantaneously. Instead, the data suggests a “multi-clump” evolution:

1. Initial Accretion: In areas of the debris disk where these are really crowded many small clumps or "moonlets" start to form at the same time.
2. Gravitational Interaction: Since these clumps are close to each other their paths start to cross and change often because of gravity pulling on them.
3. Consolidation: As time goes on these clumps bump into each other. Stick together to form bigger clumps. This “cannibalization” of smaller moonlets allows the system to consolidate its mass into one or two dominant, massive satellites.

This process is particularly relevant for the large debris fields observed in Table 4.4 the high mass availability in S1 (at grazing angle 60°) provides the “raw material” necessary for this merger process to result in a moon large enough for detection. This evolutionary pathway suggests that massive exomoons are not the result of a single “clean” formation event, but instead the survivors of a violent, competitive process within the post-impact disk.

5.2 Comparison with Previous Work

When I look at my results and compare them to what other people have found I see a few things that I agree on and a few things that I do not agree on. Most of the studies like the work by (Barr and Syal, 2017), looked at planets that are made of rock and are up to $10 M_\oplus$. My simulations went further and looked at bigger planets, up to $(20 M_\oplus)$, which is called the Giant Impactor Limit. What I found is that the rules for how moons are formed are different when we are talking about bigger planets like this. The rules, for moon formation change as a planet grows which is something that my simulations showed.

5.2.1 Efficiency of the Grazing angle

My findings strongly support the “glancing impact” theory popularized by (Barr and Syal, 2017) and (Sasaki and Barnes, 2014). They argued that oblique impacts are the only way to get enough material into orbit to form a large moon. My data shows that in S1, when the angle changed from 15° to 60° while grazing the debris mass increased a lot. This means that how the crash happens is just as important as the size of the objects, in the crash.

5.2.2 The Mass Discrepancy (Super-Earths vs. Giants)

A main point is the mass limit for making moons. (Malamud et al., 2020) found out that when planets get big and heavy it is harder for them to keep their moons because of the gravity. The mass limit for forming moons is a point of discussion. As planets get bigger the mass limit, for forming moons becomes a problem because the gravity of the planets gets stronger, and it is harder for the planets to keep the moons. My results provide direct evidence supporting this conclusion. Although S2 had more total mass, it was less efficient at producing a large debris field than the smaller S1. This contradicts earlier simplified models that assumed “more mass equals bigger moons.” Instead, my results are consistent with more recent theoretical work by (Nakajima et al., 2022), which shows that moon formation efficiency does not scale linearly with planet mass.

5.2.3 Detection Hurdles

Finally, my work adds a layer of realism to the search strategies proposed by (Kipping et al., 2012). Kipping’s “Hunt for Exomoons” project looks for objects around $0.1 M_{\oplus}$. While Heller and Barnes (2013) suggested these might be common, my data shows they are quite rare. Only my most “optimized” scenario (S1 at 60°) came close to this limit. This suggests that the “missing exomoons” in current telescope data might not be a failure of our technology, but rather a sign that the specific conditions needed to form $0.1 M_{\oplus}$ moon highly oblique impacts on mid-sized Super-Earths are rarer than we once thought.

5.3 Limitations

While my simulations provide valuable insights into the formation of exomoons around Super-Earths, there are several constraints that must be acknowledged when interpreting these results.

5.3.1 Resolution Constraints and Particle Limits

Although this study utilized a high resolution of one million particles to model the collisions, it is important to recognize that even at this scale, I am using a discrete approximation of a continuous physical event. While one million particles provide enough detail to track the primary mass distribution and the formation of large clumps, they may still reach a limit when modeling the “low-density” edges of the debris disk. In the world things are made up of tiny things like atoms and molecules; in our model, each “particle” represents a massive amount of material. The thing is, when we are talking about the system, very small things like the

turbulence in the vapor disk or the start of tiny moonlets that are just a few kilometers across might not be seen clearly in the simulation. This way of doing things is good enough to find things that are about the size of Mars or the Moon but if we use even more particles or different methods, we might be able to see the exact details of the density gradients in the disk more clearly.

5.3.2 Simplified Material Models

In my scenarios, I used standard Tillotson EoS parameters to represent the iron core and granite mantle of the planets. However, real Super-Earths can have much more complex compositions, including high-pressure ice layers or thick atmospheres. My model treats these planets as primarily "rocky," which may not account for how a thick atmosphere could create "drag" on the debris. This drag might cause more material to fall back onto the planet than my current simulations suggest.

5.3.3 Lack of Long-Term Tidal Tracking

My research focused heavily on the immediate aftermath of the impact (the first few hours to days). I found a lot of stuff in the debris field. I did not watch what happened to the system for a long time like millions of years. When we think about it the planet and the moon can affect each other with their gravity over a time. This can make the moon move closer to the planet and crash into it or move away and float off into space. So, my potential moons are a picture of what could happen not what will be there after a long time, like billions of years. The potential moons I am talking about are that potential moons they might not survive for billions of years. The planet and the potential moons and the whole system can change over time.

Chapter 6

Conclusion and Future Work

6.1 Conclusions

6.1.1 Summary of Findings

This thesis explored the complex dynamics of exomoon formation through a series of high-resolution SPH simulations, specifically looking at the differences between a $12 M_{\oplus}$ (S1) and a $20 M_{\oplus}$ (S2) system. By testing four different grazing angles, I have identified the “sweet spot” for satellite creation.

My data shows that the grazing angle is the single most important factor in determining the mass of the resulting moon. At a 15° grazing angle, both scenarios produced negligible results, yielding only Mimas-mass secondary bodies. However, as the angle increased to at grazing angle 60° , the outcomes diverged sharply. In S1, the debris field grew to a massive 4.84×10^{23} kg (comparable to Mars) in Table 4.4. In contrast, S2 reached a limit where the planet’s own gravity began to hinder the disk’s growth, resulting in a debris field nearly three times smaller than S1 at the same at grazing angle 60° .

6.1.2 Research Implications

The results of this study lead to three major conclusions for the field of planetary science:

1. The Efficiency of Smaller Super-Earths: I found out that smaller Super-Earth ($12 M_{\oplus}$), which are good at making big moons that stay in place for a long time. This is not the case for Super-Earth, like $20 M_{\oplus}$). The reason is that the big Super-Earth has a strong pull. So, when something hits this Super-Earth, most of the fragments that break off get pulled in. They do

not go into orbit around the Super-Earth. The Super-Earth just pull them back in.

2. The “Glancing Impact” Requirement: My work shows that moons, around planets or exomoons are most likely to form when things hit these planets at a small angle which is called a glancing impact. It seems to be the best way for exomoons to form around Super-Earth. Only at grazing angles of 45° and 60° I saw mass distributions that could realistically lead to the formation of detectable satellites.
3. Refinement of Search Strategies: I think the best targets for finding moons around planets are planets that are, between $10\text{--}12 M_\oplus$. These planets probably had collisions a long time ago. That is why we have not found moons yet because most surveys look at much bigger planets or Super-Earth.

6.1.3 Final Perspective

By using an iron-core and granite-mantle model with one million particles, this study has provided a high-resolution look at the violent origins of planetary systems. The way things are in the world like the air slowing things down and the gravitational pull of planets will change these moons a lot over a very long time. My results show one thing: the size of a moon is determined by how it was formed. As we get better at using our telescopes we need to focus on the ways to find moons that we identified in this study. This will be very important if we want to find the first moon, outside our system that we can confirm really exists.

6.2 Future Work

This study gives us an understanding of how exomoons form through giant impacts, but it also raises some new questions that need to be answered.

6.2.1 Inclusion of Volatiles and Atmospheric Effects

Future models should not just focus on iron and granite. Also include other materials that make up planets. For example, if we add layers of pressure ices or a thick atmosphere made of hydrogen and helium to the models, they will be more realistic for Super-Earth, like Sub Neptunes or Water Worlds. We also need to think about how the drag from the atmosphere affects the orbits of debris fields that weigh between 10^{22} kg and 10^{23} kg, like the ones I found in S1 and S2.

6.2.2 Long-term Tidal and Orbital Evolution

My simulations captured the immediate aftermath of the collision, but the long-term survival of the secondary bodies remains an open question. Future work should use N-body simulations to track the “clump” satellites found at grazing angles 45° and 60° in my simulation. This would help determine if these bodies such as the Triton-mass object in S1 eventually crash into the planet due to tidal decay or migrate outward into a stable, permanent orbit.

6.2.3 Exploring Multiple Impact Scenarios

So, I was looking at what happens when there is a crash in "Giant Impact" which tells about the history of big planets called Super-Earth it is usually a lot more complicated and often crashes. Future research could look at what happens when there are lots of crashes one, after the other, which are called impacts. This is when a second crash, which is smaller either adds to a moon that is already there or totally messes up the stuff that was left over from the first crash. This would give us an idea of how moons change over time in systems with a lot of planets close together. It would be a confusing timeline, but it would probably be more correct. The history of Super-Earth is often characterized by collisions, and these sequential impacts would help us understand this better.

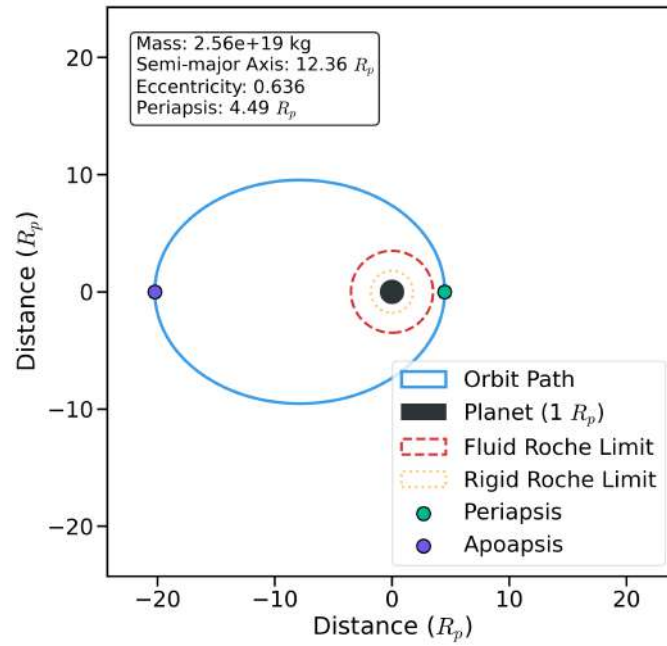
Appendix A

Additional Tables and Figures

A.1 Some Stable Orbits for S1 and S2

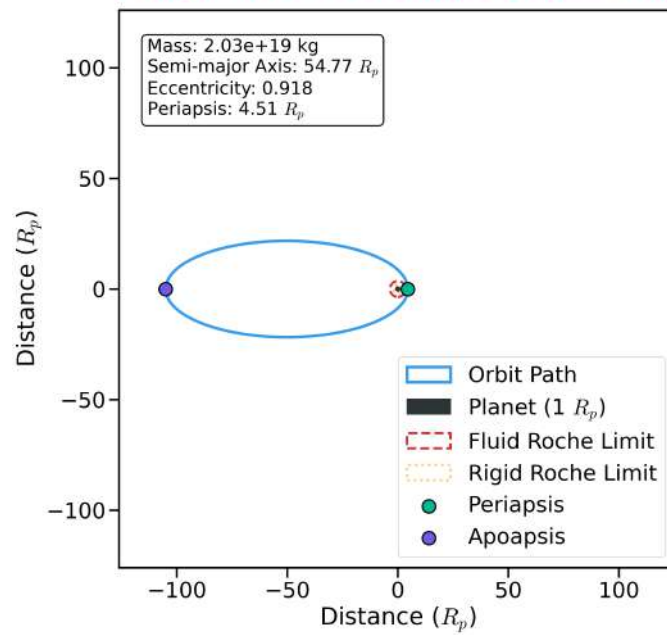
The following Figure 16 and Figure 17 illustrates that the stable captured orbits of fragments generated during tidal disruption events in S1 and S2.

Fragment 1079: STABLE (Captured)



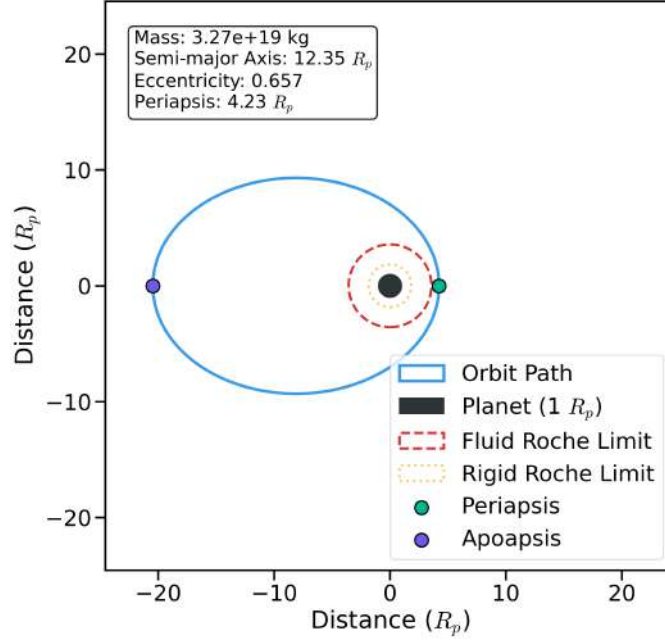
(a)

Fragment 1363: STABLE (Captured)



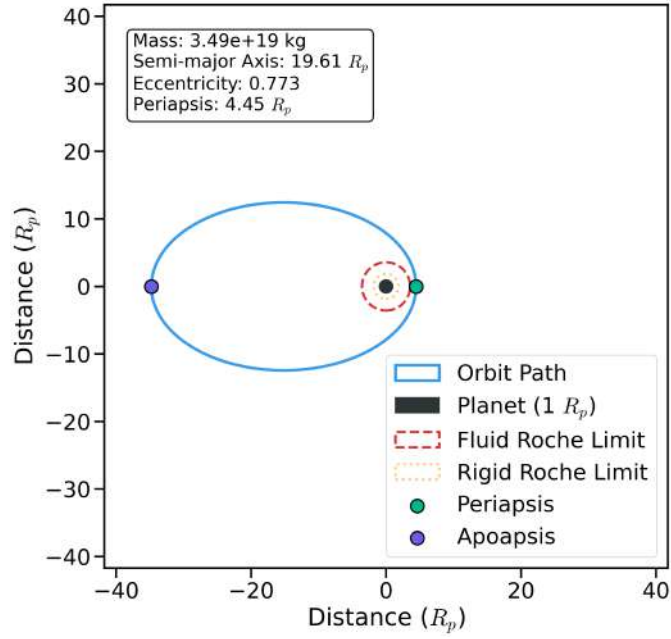
(b)

Fragment 1928: STABLE (Captured)



(c)

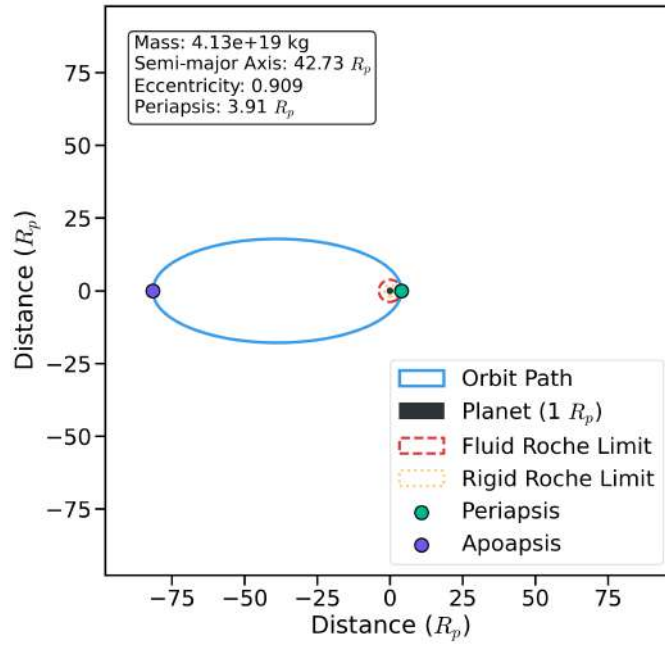
Fragment 12057: STABLE (Captured)



(d)

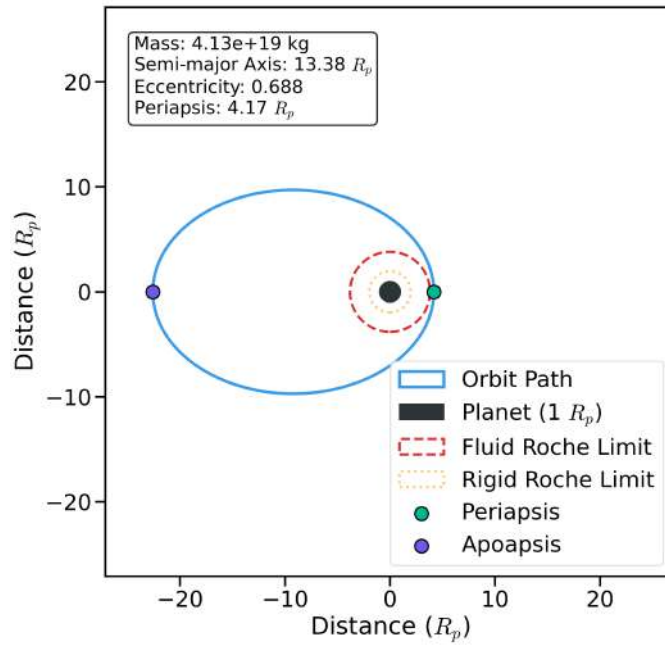
Figure 16: (a), (b), (c), and (d) show the stable orbits of fragments for grazing angles of 15° , 30° , 45° , and 60° for S1, respectively.

Fragment 254: STABLE (Captured)



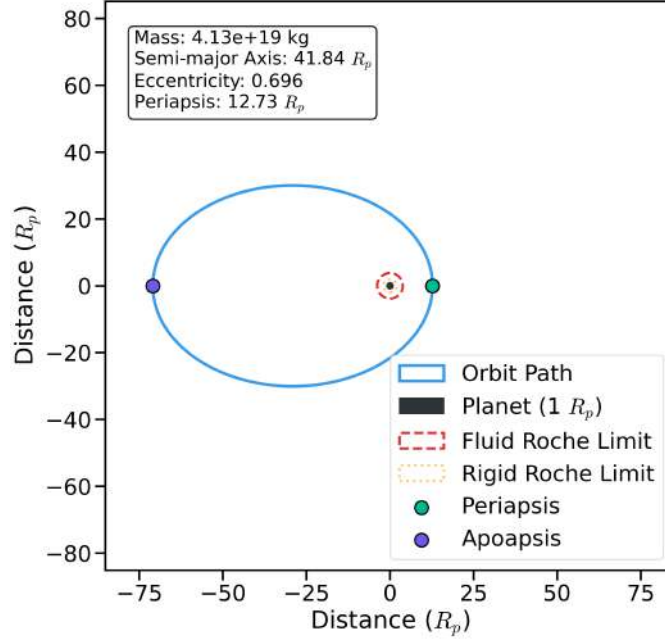
(a)

Fragment 780: STABLE (Captured)



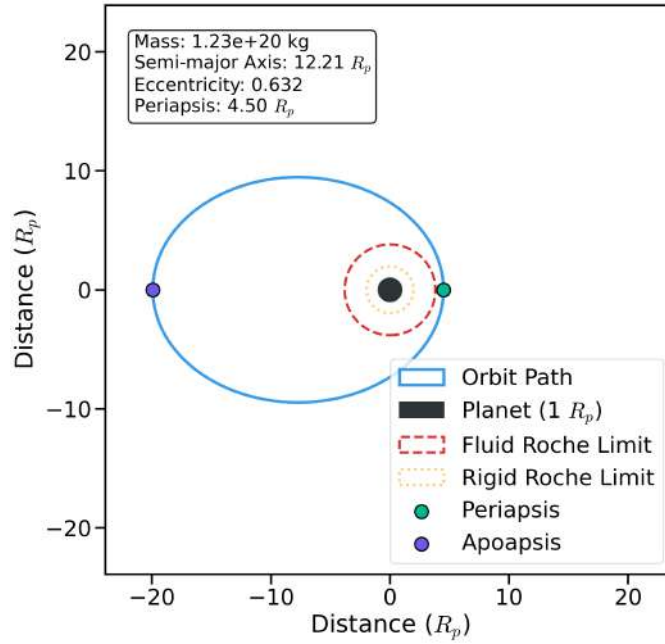
(b)

Fragment 317: STABLE (Captured)



(c)

Fragment 1011: STABLE (Captured)



(d)

Figure 17: (a), (b), (c), and (d) show stable captured orbits of fragments for grazing angles of 15° , 30° , 45° , and 60° for S2, respectively.

When we talk about the orbits of these fragments, I focus on a key thing. The major axis is one of them which is the average distance from the fragment to the planet. I also look at the shape of the orbit, which shows us how long and wide it is. Then there is the point to the planet, which is called the periapsis. The shape of the orbit is important because it helps us understand what the orbit of the fragments looks like. If the shape of the fragments orbit is, near 0 the orbit of the fragments is a circle. If the eccentricity of the fragments is near 1 the orbit of the fragments is very elliptical. The periapsis of the fragments must be outside the planet's radius. It should also be far from areas, where the tidal forces are strong so that the orbit of the fragments remains stable. I look at two tidal limits to see if the orbits are stable. The first one is the Roche limit, which is shown by a yellow dotted line. This line marks the distance inside which a rigid body could start to break because of tidal forces. The second one is the Roche limit, which is shown by a red dashed line. This line marks the distance inside which a fluid body could be completely broken apart. When I look at the figures, I notice that some fragments, like Fragment 1079 in S1 and Fragment 254 in S2 are close to these limits. The orbits of Fragment 1079 and Fragment 254 are still stable, and they move in an elliptical path, around the planet. This means that the fragments are not thrown out of the system or crash into the planet. This shows us that the orbits of the fragments are stable overall. The semi-major axis and the eccentricity of the fragments are important to understand their orbits. The fragments, such as Fragment 1079 and Fragment 254 have orbits which are stable and do not intersect with the planet, which is very important, for the fragments.

Appendix B

Mathematical Derivations

To make sense of the mass distributions observed in my SPH simulations, I apply several physical laws that dictate how debris is scattered and whether it can eventually form a stable moon. This section outlines the mathematical reasoning used to evaluate our results.

B.1 Geometry of the Impact and Angular Momentum

The final distribution of material depends heavily on the “impact parameter,” denoted as b . This value represents the offset of the collision and is directly linked to the grazing angle θ . we can calculate this as using formula:

$$b = \sin(\theta)$$

In my study the angles at which things hit are 15° , 30° , 45° and 60° . These angles correspond to b values that are between 0.26 and 0.87. When the b value is higher it means the hit is more of a glancing blow. This glancing blow is important for putting momentum into the things that are orbiting around like debris rather than, into the core of the planet. I can figure out how much angular momentum is available using the following relationship:

$$L \approx \frac{M_i + M_t}{M_i M_t} v_{\text{rel}} b (R_i + R_t)$$

My data suggests that more oblique impacts (higher b) are superior at keeping material in orbit. Without this momentum, the gravity of the primary remnant would simply reclaim the debris, preventing satellite formation.

B.2 Tidal Constraints and the Roche Limit

The moonlets need to be able to withstand the planets forces even if a lot of mass is thrown out. I use the Roche Limit to figure out the area where moonlets can come together without being ripped apart. This safe area is where moonlets can safely clump together without the planet’s forces tearing the moonlets apart. The Roche Limit is important, for determining this area for the moonlets. For a rigid body, this is defined as:

$$a_R = R_p \left(2 \frac{\rho_m}{\rho_p} \right)^{1/3}$$

In this equation, R_p is the radius of the resulting planet, ρ_p is the density of the planet and ρ_m is the density debris. In S2 ($20M_\oplus$), the immense mass and density of the primary body pushes the Roche-limit further out. This creates a larger “danger zone” where material is likely to be pulled back down, explaining why S2 struggles to form massive moons compared to the smaller S1.

B.3 Calculating Mass Partitioning Efficiency

To objectively compare how well each collision creates a potential moon, I use an efficiency coefficient, η_{disk} . This allows us to see what percentage of the total system mass (M_{tot}) ends up in the debris field (M_{debris}):

$$\eta_{\text{disk}} = \frac{M_{\text{tot}}}{M_{\text{debris}}}$$

By applying this to my simulations S1 and S2 at grazing angle 60° results, we can see a clear mathematical divide:

$$\text{Scenario 1: } \eta_{\text{disk}} \approx \frac{8.3 \times 10^{25}}{4.84 \times 10^{23}} \approx 171.48$$

$$\text{Scenario 2: } \eta_{\text{disk}} \approx \frac{1.47 \times 10^{26}}{1.51 \times 10^{23}} \approx 973.50$$

This comparison demonstrates that the $12M_\oplus$ S1 is roughly six times more efficient at generating a debris field than the $20M_\oplus$ S2 at grazing angle 60° . This mathematical proof supports my conclusion that larger Super-Earths face a “gravity trap” that hinders exomoon formation.

Appendix C

Simulation Scripts

C.1 Roche Limit Calculation

Script: roche_limit.py

```
rho_proj = core_frac_proj*rho_core + mantle_frac_proj*rho_mantle

d_roche_fluid = 2.44 * R_p * (rho_p / rho_proj)**(1/3)
d_roche_rigid = 1.26 * R_p * (rho_p / rho_proj)**(1/3)
```

C.2 Satellite Identification

Script: moon_vs_time.py

```
def get_orbital_elements(x, y, z, vx, vy, vz, central_mass):
    r = np.sqrt(x**2 + y**2 + z**2)
    v2 = vx**2 + vy**2 + vz**2
    energy = 0.5 * v2 - (G * central_mass) / r
    a = -G * central_mass / (2 * energy)
    e = np.sqrt(1 - h2 / (G * central_mass * a))
    return a, e
```

C.3 Orbital Visualization

Script: individual_orbits.py

```

semi_minor = sma * np.sqrt(1 - ecc**2)
center_x = -sma * ecc

if fragment['r_peri_m'] < ROCHE_LIMIT_RIGID:
    stability = "UNSTABLE"
elif fragment['r_peri_m'] < ROCHE_LIMIT_FLUID:
    stability = "MARGINAL"
else:
    stability = "STABLE"

ellipse = Ellipse((center_x, 0), width=2*sma,
                  height=2*semi_minor)
ax.add_patch(ellipse)

```

C.4 Radial Mass Distribution

Script: mass_distribution.py

```

def compute_center(x, y, z, m):
    M = np.sum(m)
    return np.array([np.sum(x*m)/M,
                    np.sum(y*m)/M,
                    np.sum(z*m)/M])

R = np.sqrt((x - center[0])**2 +
            (y - center[1])**2 +
            (z - center[2])**2)

mask_disk = R > R_p
hist_r, edges_r = np.histogram(
    R_disk / R_r,
    bins=np.linspace(0.1, 2.0, 121),
    weights=m_disk
)

bin_width = edges_r[1] - edges_r[0]
dM_dR_roche = (hist_r / bin_width) / np.sum(m_disk)

```

C.5 Disk Vapor Fraction

Script: disk_vapor.py

```

v2 = vx*vx + vy*vy + vz*vz
E = 0.5*v2 - G*M_planet/r
disk_mask = (E < 0) & (r > R_planet)

f_vap = (u_disk - u_iv) / (u_cv - u_iv)
f_vap[u_disk < u_iv] = 0
f_vap[(u_disk > u_cv) & (rho_disk < rho0)] = 1
f_vap = np.clip(f_vap, 0, 1)

vapor_fraction = np.sum(m_disk * f_vap) / np.sum(m_disk)

```

C.6 Particle Count Validation

Script: `particle_counts.py`

```

particle_mass = data["total_mass"] / data["n_particles"]
counts = [m / particle_mass for m in data["agg_masses"]]

```

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